Individual-specific distortion of cortical hue geometry in color vision deficiency informs personalized color correction

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Abstract: Color vision deficiency (CVD) is typically corrected with generic filters designed from a retina-based model. Such filters change how colors appear but improve color discrimination only in some products and tasks. Filter design omits visual cortex, where the retinal loss and the observer's partial compensation combine. How that combination reshapes color representation in CVD remains underinvestigated. Here we show that CVD distorts the geometry of the cortical color representation rather than weakening its overall signal. We scanned two adults with CVD, one with each red–green subtype, and seven healthy controls. All eight colors remained decodable from cortical activity in both CVD participants, whereas the continuous hue geometry departed from controls, differently in each individual. We modeled each distortion as a hue rotation along two color-space axes and inverted the fit into a personalized stimulus-space filter. Both CVD participants completed a second session of fMRI and behavioral testing to evaluate the filters. Under the individualized filter, the four hue-discrimination thresholds that had exceeded the control range in the first session (three for deutan, one marginal for protan) moved within that range. This small sample cannot establish superiority over a deployed accessibility filter. The direction of cortical change differed across participants and measures, and neither reached the healthy reference. This study identifies a previously uncharacterized geometric distortion of the cortical color representation in CVD and introduces, to our knowledge, the first cortically grounded framework for individualized color correction. Systematic studies can quantify these distortions and provide a new class of personalized filters.

1 Introduction

Color vision deficiency (CVD) affects approximately 8% of men and 0.4% of women of European ancestry (Birch, 2012). It arises from a shift in the spectral sensitivity of the L- or M-cone photopigment. The magnitude of that shift varies continuously across affected individuals (Bosten, 2019). X-linked polymorphisms in the cone opsin genes underlie the shift (Deeb, 2005; Neitz & Neitz, 2011). The two pigments lie between 1 and 12 nm apart in anomalous trichromacy, compared with roughly 27 nm in normal trichromacy (Bosten, 2019; Neitz & Neitz, 2011). This narrowed separation attenuates the L–M cone-opponent signal without abolishing it. Red–green CVD comprises two subtypes that differ in the direction of the shift. In protan observers the L-cone sensitivity moves toward the M cone, and in deutan observers the M-cone sensitivity moves toward the L cone (Neitz & Neitz, 2011).

Observers who share a diagnostic label differ substantially in how much their color space is compressed. Anomalous trichromats show reduced red–green discrimination, frequent category confusions, and elevated just-noticeable differences along the L–M chromatic axis. The severity of these signs extends from near-normal to near-dichromatic within one diagnostic subtype (Bosten, 2019; Neitz & Neitz, 2011).

Current CVD correction is calibrated to a population-average retina and applies the same transform to every user. Hardware filters are notch lenses. Each notch attenuates a narrow band of the spectrum where the L and M cones overlap. Removing that band widens the difference between the two cone signals (Álvaro et al., 2022). The EnChroma lens cuts three such notches at wavelengths taken from a normal retina (Gómez-Robledo et al., 2018). The lens changes appearance without changing performance. Colors look more saturated and further apart along the red–green axis, while discrimination thresholds move minimally (Somers et al., 2024). The lens produces no clinically meaningful improvement on plate, arrangement, discrimination, naming or sorting tasks, and does not change diagnostic classification (Álvaro et al., 2022; Gómez-Robledo et al., 2018). Two commercial lenses shifted discrimination thresholds in 51 individuals with red–green CVD. For both lenses, variation between observers exceeded the mean effect (Patterson et al., 2022). Software filters start from the same population-average retina.

They forward-project a stimulus onto a deficient retina, either by the Brettel–Viénot–Mollon construction for dichromacy (Brettel et al., 1997) or by the parameterized Machado model, which extends the same logic to anomalous trichromacy (Machado et al., 2009). Daltonization inverts these models. It recolors an image, moving red–green differences onto the blue–yellow and lightness axes (Fidaner et al., 2005; Simon-Liedtke & Farup, 2016).

Individualized correction has been implemented at the retinal level. It tunes a cone-shift parameter to a user's responses on plate-style or appearance-matching tests. That parameter can be estimated accurately, from opsin genotype or from cone spectral sensitivities measured in observers of known genotype (Deeb, 2005; Stockman & Sharpe, 2000). The retinal parameter predicts color discrimination imperfectly. Some anomalous trichromats with very small spectral separations discriminate better than their separation predicts (Neitz & Neitz, 2011). Across observers, the correlation between cone spectral separation and color discrimination is weaker than expected (Bosten, 2019). A retinal parameter fixes the input to the visual system. It does not describe the cortical representation that is built from that input and determines color perception. Filters tuned to an observer's own cone sensitivities trade one set of confusions for another rather than moving color vision toward normal (Gómez-Robledo et al., 2018). The remaining step is a filter derived from the user's own cortical color representation.

Cortical activity patterns specify both which color was seen and where that color lies among the others. Cortical color processing is distributed and hierarchical (Conway et al., 2018; Gegenfurtner, 2003; Shapley & Hawken, 2011). Early visual cortex carries cone-opponent signals. In V1, perceptual hues form spatially clustered response patterns, while the cardinal directions do not (Parkes et al., 2009). V1 responses are also selective for hues that fall between the cone-opponent axes (Kuriki et al., 2015). In hV4, a hue withheld from training can be reconstructed from the responses to its neighbors (Brouwer & Heeger, 2009). The relative positions of colors in these response patterns constitute a geometry, and a color can be identified in one observer from the response patterns of others (Bannert & Bartels, 2025).

What has been measured in CVD is the strength of the color signal. Anomalous trichromats perceive more red–green contrast than a cone-sensitivity account predicts (Boehm et al., 2014). Deutan observers adapt to chromatic contrast more than the same account predicts (Basim et al., 2025). Both results are consistent with a post-receptoral gain that magnifies the weakened L–M signal. Long-term adaptation to the altered input can produce such an adjustment (Webster, 2015). In V1, color responses are weaker in anomalous trichromats than in normal trichromats. In ventral V2 and V3 they are indistinguishable (Tregillus et al., 2021).

Color deficiency also changes where colors lie relative to one another. Anomalous trichromats still describe stimuli with all four hue primaries, but each primary lands on a different stimulus than it does for normal trichromats (Emery et al., 2021). When dichromats judge the difference between color samples, red and green fall close together, and the map recovered from those judgments has one axis where normal trichromats have two (Saysani et al., 2018). When protan and deutan observers judge the difference between Munsell chips, the map keeps two axes but is dented at yellow and purple-blue. The size of that dent varies more than tenfold across observers, and some show no dent at all (Ohkoba et al., 2022). All of these are reports. The same relative positions have yet to be measured in the cortical response.

We take the individual's own cortical color representation as the reference and invert it into a correction on the stimulus. Of these regions, hV4 is where the arrangement of responses follows perceptual color space (Brouwer & Heeger, 2009). Our design adopts the premise that shifting an individual's hV4 color geometry toward that of the healthy controls moves their color perception with it.

Three gaps remain. *(Gap 1)* The cortical color geometry of an individual with CVD remains to be described. Existing descriptions come from healthy observers (Brouwer & Heeger, 2009, 2013; Kuriki et al., 2015). *(Gap 2)* The distances between colors have not been measured in the cortical response in CVD. fMRI in CVD has reported response magnitude, including contrast-response gain in anomalous trichromacy (Tregillus et al., 2021) and activation in one dichromat and two observers with achromatopsia (Rina, 2024). *(Gap 3)* A filter inverted from an individual's own cortical representation remains underinvestigated. Existing designs invert models of the retina (Brettel et al., 1997; Machado et al., 2009) or maps of an individual's color judgments (Ohkoba et al., 2022).

We ask whether an individual's own cortical color representation can be turned into a correction for that person. We characterize two adults with CVD, one deutan and one protan, against a reference group of healthy controls (see Methods).

A stimulus-space filter is warranted and tractable in one specific regime, defined by two conditions. The displayed colors must remain decodable from the cortical response. This preserved decodability gives the filter a signal to act on. The continuous arrangement of hues must be distorted, so that there is something to correct. Our questions follow this logic.

1. **Describe.** Does an individual's cortical hue geometry in CVD carry structured distortion, with the individual hues still decodable and their continuous arrangement impaired?

2. **Summarize.** Can a few interpretable parameters capture that distortion, treated as a compact description rather than as physiological measurements?

3. **Correct.** Can a realizable, individualized filter be derived from those parameters?

4. **Validate.** Does the filter move the cortical representation closer to the healthy-control reference than a deployed accessibility filter does?

We expect the continuous arrangement of hues to be selectively impaired while the individual hues remain decodable. A few interpretable parameters should capture this distortion, and inverting them should yield a realizable, individualized correction.

Together, these steps yield a filter built for one individual. Its input is the cortical signal, its parameters are the fitted description, and its output is a stimulus shift. Each is defined at the level of the single observer, the level at which any correction must ultimately act. This study tests the feasibility of the inversion, namely whether an individual's cortical color geometry can be turned into a realizable filter. It does not test whether per-person correction outperforms a subtype average, which would require several individuals within one subtype.

2 Methods

The analysis proceeded in three stages, summarized in Figure 1C. First, we characterized each CVD participant's cortical color representation with three neural analyses. Leave-one-run-out (LORO) decoding tested whether color classification remained intact (§2.8). Leave-one-color-out (LOCO) decoding tested whether continuous hue interpolation was impaired and yielded a per-hue vulnerability profile (§2.8). Procrustes disparity quantified how far each CVD hue geometry departed from the healthy-control reference (§2.9). Second, we fitted two classes of CVD distortion model by minimizing a combined neural and psychophysical loss. Parameters were selected by a pre-specified three-gate procedure (§2.10–§2.12). Finally, we inverted each fitted distortion numerically to obtain a per-subject stimulus-space correction filter (§2.14). Figure 3 summarizes the fitting and inversion procedure. Each filter was evaluated in a second session that combined fMRI with the two psychophysical tasks described in §2.3.

2.1 Participants

Ten volunteers provided written informed consent and were enrolled; three further respondents were excluded at pre-consent screening for not meeting MRI safety criteria. The study was approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023) and conducted in accordance with the Declaration of Helsinki. All participants reported normal or corrected-to-normal visual acuity. Color vision was screened with the 14-plate Ishihara pseudoisochromatic test (Ishihara,

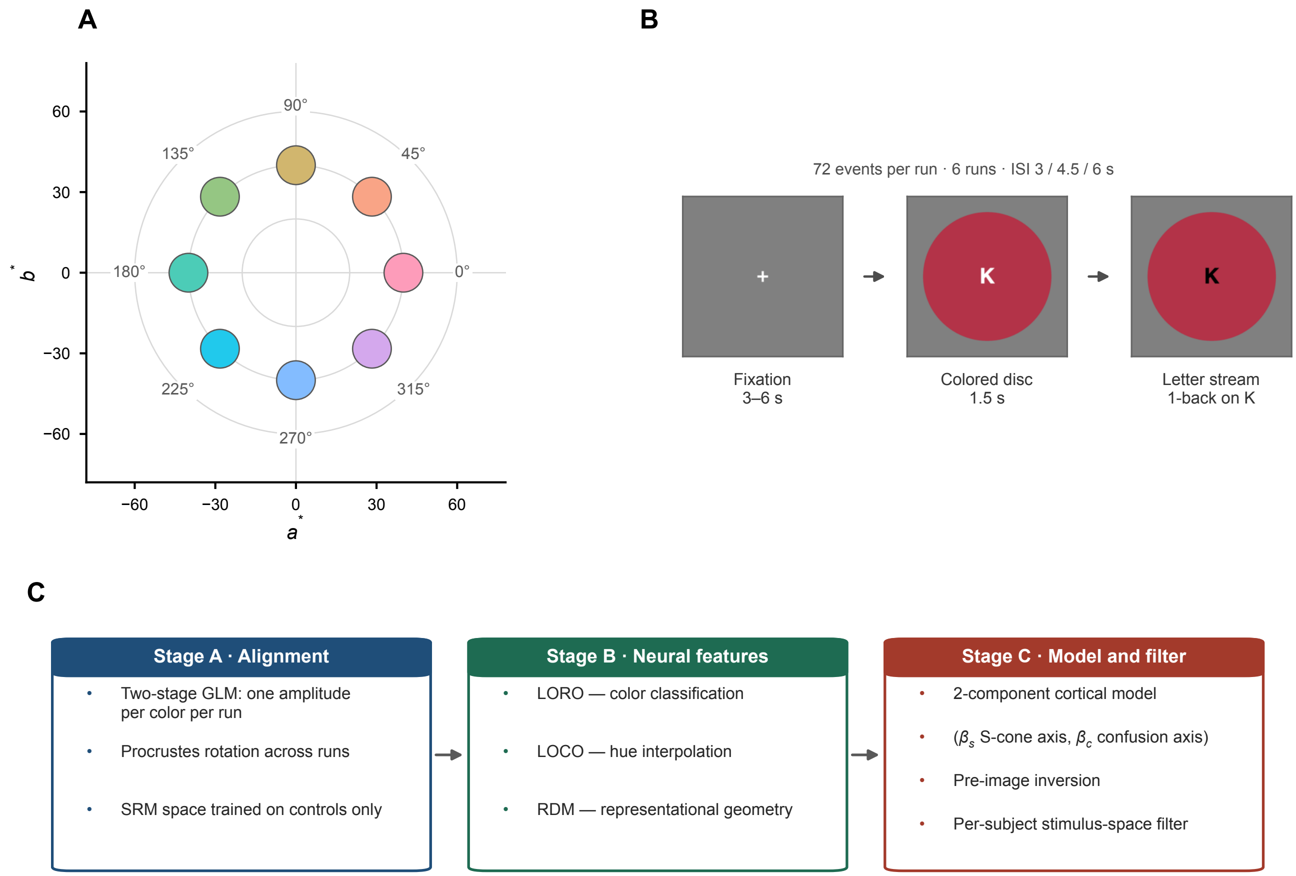

Figure 1: **Stimuli, task, and analysis pipeline. (A)** Stimulus set: 8 CIE $L^*a^*b^*$ hues at matched lightness, 45° angular spacing ($L^* = 75$, chroma $= 40$; red–magenta), shown in the a^*b^* plane. **(B)** Task: rapid serial visual presentation (RSVP). A fixation period (randomized 3–6 s) is followed by a 1.5 s uniform colored disc; 5–9 trials per color per run (Neurodesign-optimized allocation, identical across participants), six runs. **(C)** Analysis pipeline. Stage A (Alignment): two-stage GLM per-color amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Stage B (Neural feature): LORO classification, LOCO interpolation, and the SRM-space representational dissimilarity matrix (RDM). Stage C (Modeling & filter): a 2-component cortical model with S-cone-axis (β_s) and confusion-axis (β_c) components; its pre-image yields the per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07).

1917). Participants who identified 9 or more plates correctly were classified as healthy controls (HC, $n = 7$, 3 female, aged 20–27 years, mean 22.7 ± 2.5). Participants scoring below this threshold were classified as CVD. The classification plates of the same test distinguished the protan from the deutan subtype and indicated anomalous trichromacy rather than dichromacy in both. Two CVD participants completed both sessions, both male. The deutan participant was 24 years old and identified 5 of 14 plates correctly. The protan participant was 25 years old and identified 7 of 14 plates correctly. One further enrolled CVD participant completed the first session but became unavailable before the filter-evaluation session and was excluded from all analyses, leaving nine participants in the analyses. All CVD results are reported as single-case comparisons against the HC distribution with the Crawford and Howell (1998) modified t-test. No population-level inference is made.

2.2 fMRI stimuli and task

Eight chromatic stimuli were defined in CIE $L^*a^*b^*$ space at equal angular spacing in the a^*b^* plane, from $0°$ to $315°$ in $45°$ steps (Commission Internationale de L'Eclairage, 1986). All eight shared a lightness of $L^* = 75$ and a chroma of 40, and they spanned red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler at the same lightness with $a^* = b^* = 0$ served as a null condition. Isoluminance was not determined for individual participants. Stimuli were presented on a BOLDscreen 32 UHD monitor (Cambridge Research Systems) at 3840×2160 resolution, which combines a wide-gamut LED backlight with manufacturer-integrated calibration. The display was used at its factory calibration without further colorimetric verification in the scanner environment. Stimulus size is reported in display pixels throughout, since both presentation programs addressed their display in pixel units.

Stimuli were presented in a rapid serial visual presentation (RSVP) design to sustain attention and fixation. Each trial displayed one uniform chromatic disc, 500 pixels in diameter, for 1.5 s in a pseudo-random order optimized with Neurodesign (Durnez et al., 2018). A rapid serial letter stream ran at the center of the disc, and participants pressed a key on an MRI-compatible button box whenever the letter K appeared twice in succession. Inter-stimulus intervals were drawn from $\{3, 4.5, 6\}$ s. Each run comprised 72 events and lasted approximately 7 min, and each session comprised six runs. The experiment was implemented in PsychoPy 2022.2.5 (Peirce et al., 2019).

The schedule was optimized for detection power rather than for equal counts, so color counts varied within a run. Each chromatic stimulus occurred 5–9 times per run and 38–47 times across the six runs, and the remaining 14–20 events per run presented the gray filler. Every participant received the same optimized schedule, so this allocation contributes a common component to all individual estimates. The general linear model estimated one amplitude per color and run regardless of how many trials contributed, so all downstream analyses operated on a balanced 6 (run) $\times$ 8 (color) design.

2.3 Psychophysical tasks

All participants completed two psychophysical tasks on the day of the first scanning session. The first measured just-noticeable differences (JND) in hue discrimination, and the second was an eight-alternative forced-choice (8AFC) color-identification task. Both CVD participants repeated both tasks under each filter in the second session (§2.15). Both tasks ran outside the scanner under normal room lighting, on a built-in Liquid Retina XDR display set to the Apple XDR Display color preset, which was also the preset used to drive the in-scanner display.

Just-noticeable difference (JND). On each trial two chromatic discs appeared simultaneously to the left and right of fixation, and participants judged whether the two were the same or different. Each disc had a radius of 96 pixels in a 1024×768 window, their centers were separated by 260 pixels, and the side of the reference disc was randomized across trials. A fixation cross preceded each trial for 0.5 s, and the discs remained on screen until the participant responded. The two hues were separated by a fraction t of the pair's full separation, interpolated in CIE LCh, and t served as the staircase level. Two staircases per pair were interleaved from starting levels $t = 0.8$ and $t = 0.5$, both following a 1-up/1-down rule that converges on the 50% "different" point (Levitt, 1971). The step size was 0.15 for the first two reversals and 0.03 thereafter. A staircase terminated at eight reversals and at least 20 trials, extended by two further reversals whenever the last six reversal levels had an SD above 0.10. The threshold was the mean of the last six reversal levels obtained after the first two. The two discs always differed, so the procedure included no same-trials and the resulting threshold is not corrected for response bias.

Eight hue pairs were selected to span the S-cone axis and each subtype's confusion axis. Yellow–purple lies along the S-cone axis, and blue–purple and red–orange flank it. Orange–yellow and yellow–green bracket the deutan confusion axis, and green–blue and cyan–magenta bracket the protan confusion axis. Red–cyan, separated by $180°$, served as a control against ceiling effects. Pair selection was fixed on these grounds before any threshold was collected. For each pair we computed the JND ratio $\gamma = \gamma_{\mathrm{CVD}}/\bar{\gamma}_{\mathrm{HC}}$, in which $\bar{\gamma}_{\mathrm{HC}}$ denotes the HC mean threshold. Ratios above 1 indicate elevated hue-discrimination difficulty relative to HC. Because all participants were measured on the same display, the ratio is unaffected by any difference between that display and the one used in the scanner. These ratios entered the fitting loss described in §2.11.

8AFC color identification. Each trial began with a 1.0 s fixation, followed by a uniform chromatic disc of 500 pixels in diameter for 1.5 s and a 0.5 s blank interval. Participants then selected the presented hue from the eight scanner stimulus colors, displayed as a two-by-four grid of discs 100 pixels in radius at 240-pixel horizontal and vertical spacing, and navigated with the arrow keys. The positions of the eight options were reshuffled on every trial. A trial was recorded as a timeout if no response was given within 10 s, and no feedback followed. Each run comprised 64 trials, eight per color, in random order. Identification accuracy served as an independent psychophysical readout, held out from the fitting loss.

2.4 MRI acquisition and preprocessing

Imaging was performed on a Siemens 3T MAGNETOM Cima.X scanner with a 64-channel head coil. A whole-brain T1-weighted MPRAGE volume served as the anatomical reference. It was acquired sagittally at 1 mm isotropic resolution, with a repetition time of 2000 ms, an echo time of 2.36 ms, a field of view of 256 mm, a bandwidth of 220 Hz per pixel, and GRAPPA acceleration factor 2. Functional coverage was restricted to posterior occipital cortex with 24 interleaved oblique slices aligned perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009), spanning 48 mm. The functional sequence used a repetition time of 1.5 s, an echo time of 30 ms, a flip angle of $70°$, and 2 mm isotropic voxels over a 192 mm field of view on a 96×80 matrix. Acceleration was GRAPPA factor 2 without multiband, the bandwidth was 1447 Hz per pixel, and phase encoding ran right to left. Each run yielded 288–292 volumes. A gradient-echo field map with echo times of 3.88 and 6.34 ms was acquired in each session and associated with the functional runs in the BIDS metadata.

Data from the first session were converted to BIDS format with the ezBIDS web service (Levitas et al., 2024), accessed in October 2025, which also defaced the anatomical image. The second session was converted with dcm2bids 3.2.0 (Boré et al., 2024) in June 2026 and was not defaced. Both routes used `dcm2niix` for DICOM conversion. Preprocessing followed a custom pipeline rather than a standard container, and no confound regression was applied at any stage (Supplementary §S1). All steps used FSL 6.0.5.1 and FreeSurfer 7.2.0. The T1w image was skull-stripped with `bet2` (Smith, 2002), and functional images were registered to it by mutual-information coregistration initialized from the scanner header, using `mri_coreg` (Maes et al., 1997; Wells et al., 1996). Normalization to the ICBM 152 Nonlinear Asymmetric 2009c template (MNI152NLin2009cAsym) used a twelve-parameter affine transform (FLIRT) followed by nonlinear warping (FNIRT) (Andersson et al., 2007; Jenkinson et al., 2002), yielding 2 mm isotropic BOLD data in MNI space. Each functional volume was resampled once by this composed transform, without slice-timing correction, motion realignment, susceptibility distortion correction, or spatial smoothing. Head motion was estimated separately by rigid-body registration of every volume to its run's middle volume, using MCFLIRT (Jenkinson et al., 2002), and retained as a quality-control record. Mean framewise displacement was 0.32 ± 0.04 mm across the nine analyzed participants (HC 0.31 ± 0.04, CVD 0.34 ± 0.05), with 16.2% of volumes exceeding 0.5 mm (Supplementary §S1). Because the primary analysis left head motion uncorrected, every neural endpoint was recomputed with the motion parameters and their temporal derivatives entered as nuisance regressors (Supplementary §S2).

2.5 ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the probabilistic atlas of visual topography of Wang et al. (2015), resampled to 2 mm and thresholded at a probability above 50%. V1, V2, and V3 each merge the ventral and dorsal subdivisions of the atlas. The threshold was applied to each ROI independently, so a voxel assigned to more than one ROI was retained in each. Each ROI was then intersected with that participant's BOLD brain mask. Mean voxel counts across the nine analyzed participants were 665 for V1, 458 for V2, 105 for V3, and 64 for hV4, with standard deviations of 209, 113, 19, and 18. No further functional voxel selection was applied.

Response amplitudes were estimated with a two-stage general linear model (Brouwer & Heeger, 2009, 2013; Dale, 1999). In the first stage a finite impulse response model with eight delays, spanning 0 to 10.5 s, was fitted to the concatenated runs without drift terms. Averaging the fitted responses over voxels and over the eight colors gave one hemodynamic response function per ROI and participant. In the second stage that function and its temporal derivative were convolved with per-color event sequences and fitted to each run separately by ordinary least squares, alongside a linear and a constant drift term for that run. The result was one amplitude for every voxel, color, and run, arranged as a $6 \times 8 \times V$ array per participant.

Amplitudes were aligned across runs by a within-subject Procrustes rotation (Gower, 1975). For each of runs 2 to 6, the 8×8 orthogonal matrix Q minimizing $\|X_1 - X_r Q\|_F$ was applied to that run's color-by-voxel matrix X_r, where X_1 is run 1. Scaling was not permitted, so the rotation acts within the eight-dimensional space of color responses and leaves each voxel's identity unchanged.

2.6 Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i^{\mathrm{srm}} \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i^{\text{srm}}, S} \sum_{i=1}^{N} \|X_i - W_i^{\text{srm}} S\|_F^2 \quad \text{s.t.} \quad W_i^{\text{srm}\top} W_i^{\text{srm}} = I_k \tag{1}$$

The orthonormality constraint makes the mapping from the shared space back into each participant's voxel space an isometry. Distances and angles in the shared space are therefore those of the reconstructed color patterns. SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W^{\text{srm}}_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. The reduced dimension was chosen by seven-fold leave-one-subject-out cross-validation, withholding one HC participant from SRM training on each fold. Candidate values from 2 to 6 were ranked on each fold by three quality metrics, namely RDM split-half reliability, cross-subject RDM correlation, and reconstruction error, and the value with the best mean rank was selected. This gave $k = 4$ for V1 and V2 and $k = 3$ for V3 and hV4.

2.7 Forward encoding model

Both classification (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a $F = 6$-dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \ldots, c_F(\theta)]^\top$, where $c_k(\theta) = \max\big(0, \cos(\theta - \theta_k)\big)^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k, with preferred hues spaced equally at 60°. Stacking the eight stimulus channel vectors row-wise gives the channel design matrix $C \in \mathbb{R}^{8\times F}$. Decoding and encoding optimize different targets and therefore use different estimators of the channel-to-voxel weight matrix $W \in \mathbb{R}^{F\times V}$, which is distinct from the SRM mapping W^{srm}.

Decoding (stimulus hue from voxels). Decoding recovers the stimulus hue from voxel responses. We estimated W by ordinary least squares, taking the pseudoinverse of the channel design matrix C. For a held-out voxel pattern $\mathbf{x}$, channel activations were reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$. The decoded hue was the angle, among all 360 integer hues, whose basis vector correlated most strongly with $\hat{\mathbf{c}}$ (Brouwer and Heeger, 2009; Figure 2). For eight-way classification accuracy this angle was assigned to its nearest stimulus hue. This correlation readout was carried over from the forward model and was subsequently compared against five alternative decoders, which is reported in Supplementary §S10.

Encoding (voxel responses from channels). Encoding predicts the voxel responses from the channels as $\hat{X} = CW$, where $\hat{X} \in \mathbb{R}^{8\times V}$. We estimated W by ridge regression. A single penalty α was chosen for each fit by generalized cross-validation (Golub et al., 1979), minimizing the GCV score averaged over the voxels of the ROI (Supplementary §S7). Regularization is warranted because the few stimuli and the correlated basis channels would otherwise overfit W and degrade prediction of held-out responses (Kay et al., 2008; Naselaris et al., 2011). This ridge prior is isotropic and imposes no spatial structure on the voxels.

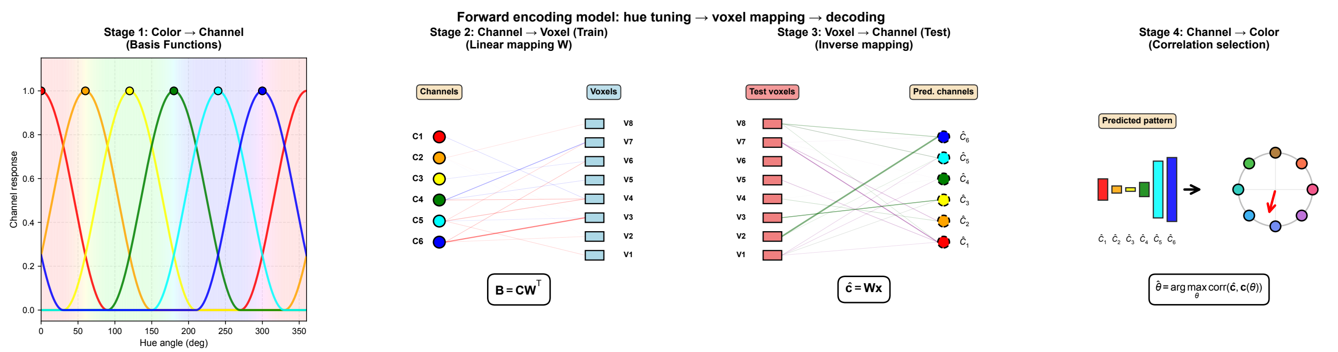

Figure 2: **Forward encoding model schematic.** Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): the channel-to-voxel weights W are fit by ordinary least squares for hue decoding, or by GCV-regularized ridge regression for voxel-response prediction (encoding). Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$. Stage 4: the decoded hue is recovered as the angle, among all 360 integer hues, whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same model serves both decoding schemes (Section 2.8): cross-run classification (LORO) and cross-color interpolation (LOCO). Their basis and correlation-based selection are identical, and only the held-out test set differs. Decoding uses the OLS pseudoinverse weights.

2.8 Two decoding schemes: cross-run classification and cross-color interpolation

Two complementary cross-validation schemes applied the forward encoding model to characterize each participant's hue processing. Leave-one-run-out (LORO) removed one run at a time. Every hue therefore remained in the training set, and the scheme tested the categorical representation of the eight hues, measured as color classification. Leave-one-color-out (LOCO) removed all six runs of one hue at a time. The held-out hue therefore never appeared in training, and the scheme tested the continuous representation, measured as hue interpolation. Both schemes were computed on the Procrustes-aligned amplitudes of §2.5.

Cross-run classification (LORO) Color classification was assessed by leave-one-run-out cross-validation. For each of the six runs, W was estimated on the remaining five runs from mean-subtracted amplitudes, and the eight color patterns of the held-out run were decoded by the correlation readout of §2.7. Primary performance was the eight-way classification accuracy averaged across the six folds, against a chance level of 0.125. Single-case comparisons against the healthy controls used the Crawford and Howell (1998) modified t-test. The test is two-tailed here because the hypothesis is preservation, whereas the interpolation tests below are one-tailed.

Cross-subject transfer of the classifier was evaluated separately. All eight hues enter training here, so this analysis extends LORO across participants and has no interpolation counterpart. The encoder W maps the six channels onto the voxels of the participant it was fitted to. Applying it to a second participant therefore requires the two voxel spaces to correspond, which the SRM-aligned space supplies. Transfer to a held-out control used a leave-one-control-out encoder trained on the remaining six controls, giving 28 participant-by-ROI cells. Transfer to CVD used an encoder trained on all seven controls, giving 8 cells. The two destinations were compared by a Mann–Whitney U test with rank-biserial correlation as the effect size. Cells from the same participant are not independent, so this comparison is descriptive.

Cross-color interpolation (LOCO) Continuous hue interpolation was assessed by leave-one-color-out cross-validation. Each of the eight hues was held out in turn. W was estimated from the remaining seven hues across all six runs, pooled into a 42-sample design matrix, and the held-out hue was predicted by the same correlation readout.

The primary ROI for interpolation was hV4. Brouwer and Heeger (2009) reconstructed novel colors from V4 and VO1, the ROIs where their interpolation was strongest. LOCO was computed in all four ROIs. Crediting a ROI with interpolation required the healthy-control mean to exceed the color-label permutation null defined below. This criterion tests specificity to color identity rather than overall decodability. It also fixes where a CVD reduction is interpretable, since the comparison rests on controls who interpolate above the null. Single-case CVD comparisons are reported for ROIs meeting the criterion. Values for all four ROIs are given in the Results.

Adjacent accuracy The primary cross-color decoding metric was adjacent accuracy, the proportion of predictions falling within $45°$ of the target hue. This tolerance corresponds to one hue step. Values were averaged across the six runs of each test color and range from 0 to 1. The readout selects among all 360 integer hues, so a prediction drawn uniformly from that output space falls within tolerance on 91 of 360 draws, giving a chance level of 0.25. Exact accuracy, which requires the prediction to fall in the correct one of eight $45°$ bins, has a chance level of 0.125. Healthy-control performance was tested against an empirical null of 1,000 random color-label permutations rather than against the analytic chance level. Single-case HC–CVD comparisons at hV4 used the Crawford and Howell (1998) modified t-test, one-tailed and lower.

Vulnerability profile The eight-dimensional vector of per-hue adjacent-accuracy values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant's *hue vulnerability profile*. For group-level model fit, we report Spearman ρ between the observed $\mathbf{v}$ and the $\mathbf{v}$ predicted by the CVD distortion model (see the Candidate CVD distortion models section). Per-hue group comparisons (Figure 4C) used the Crawford and Howell (1998) modified t-test, one-tailed and uncorrected. Effect sizes are reported as the case-control index $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}}$, which equals $t\sqrt{(n+1)/n}$ (Supplementary §S21).

2.9 Representational geometry

Procrustes disparity measures how far a participant's whole representational geometry lies from the healthy-control reference in SRM-aligned space. Disparity was the Procrustes distance between a participant's SRM-aligned $8 \times k$ pattern and that reference. Both patterns were centered on their column means and scaled to unit Frobenius norm. Disparity was then the residual Frobenius norm after the optimal orthogonal rotation.

The HC disparity distribution was estimated by a leave-one-HC-out procedure. Each HC participant was compared to the mean of the remaining six, which keeps that participant out of its own reference. Individual CVD disparity was compared to this distribution by the Crawford and Howell (1998) modified t-test, one-tailed upper at $\alpha = 0.05$. The direction follows the prediction that CVD geometry lies farther from the HC reference than HC geometry does.

Two estimates of the distribution were computed (Supplementary §S12). In the first, the shared SRM space is trained on all seven HC, so each HC reference is in-sample while the projected CVD participant is out-of-sample. In the second, the space is retrained on six HC at each fold, and the held-out HC is projected by SVD on the same terms as each CVD participant. Each CVD participant receives the mean of its seven fold disparities, whereas each HC participant contributes the single value from the fold that held it out. The control distribution therefore retains fold-to-fold variance that the CVD statistic averages out, which makes the comparison conservative.

The two estimates serve different purposes. Every other stage of the pipeline operates in the all-HC space, so that space defines the representation the filter is fitted in. Testing whether a participant departs from the healthy-control distribution requires the held-out participant to be projected on the same terms as the CVD participants, which the symmetric estimate provides. We report both and treat the symmetric estimate as the inferential test (table S5). Because both patterns are scaled to unit norm before the rotation, disparity is insensitive to any difference in overall response magnitude, which is examined separately (Supplementary §S5).

2.10 Candidate CVD distortion models

Two candidate model classes were evaluated. Each parameterizes the angular distortion $\delta\theta(\theta)$ that maps the displayed hue angle θ to the angle perceived by a CVD observer.

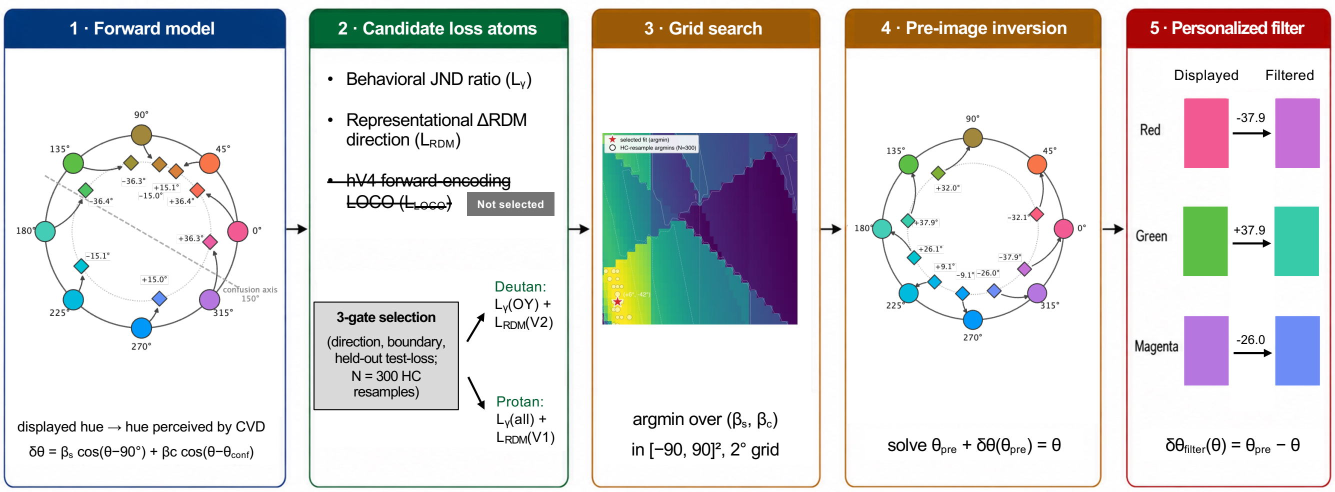

Figure 3: **Fitting a per-subject cortical distortion and inverting it into a stimulus-space filter.** **(1)** A two-parameter forward model expresses the per-hue distortion as $\delta\theta(\theta) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\mathrm{conf}})$: an S-cone-axis component (β_s) plus a family-specific confusion-axis rotation (β_c; $\theta_{\mathrm{conf}} = 16^\circ$ protan, 150° deutan). Outer ring, displayed hue; inner ring, hue perceived under the fitted distortion; labels give the signed displacement. **(2)** Three candidate loss atoms quantify CVD–HC differences. A three-gate procedure (direction, boundary saturation, held-out test-loss over $N = 300$ HC resamples; §2.12) selected a different combination for each participant (deutan, $\gamma_{\mathrm{OY}} + L^{(V2)}_{\mathrm{RDM}}$; protan, $\gamma_{\mathrm{all}} + L^{(V1)}_{\mathrm{RDM}}$); the LOCO atom entered neither selected combination. **(3)** A grid search over $(\beta_s, \beta_c) \in [-90^\circ, 90^\circ]^2$ at 2° resolution minimizes the z-scored composite to the selected fit $\hat{\beta}$. The surface shown is the deutan participant's; both landscapes and their resample uncertainty are in Figure 6. **(4)** The fitted forward map is numerically inverted: for each displayed hue θ, the corrected stimulus θ_{pre} solves $\theta_{\mathrm{pre}} + \delta\theta(\theta_{\mathrm{pre}}) = \theta$. **(5)** The individualized filter is $\delta\theta_{\mathrm{filter}}(\theta) = \theta_{\mathrm{pre}} - \theta$; three of the eight hues are shown for the deutan participant, and the full eight-hue filter for both participants is in Figure 7. Hue wheels and swatches are computed from each participant's fitted parameters; the schematic reports no new statistics.

Candidate Model 1: retinal-plus-gain (R+C). The prevailing account attributes CVD color distortion to a cone-spectral shift at the retinal level (Machado et al., 2009), optionally augmented by a cortical compensation gain g (Boehm et al., 2014; Tregillus et al., 2021):

$$\delta\theta_{\mathrm{RC}}(\theta;\, g) = (2 - g)\, \delta\theta_{\mathrm{Mach}}(\theta) \tag{2}$$

where $\delta\theta_{\mathrm{Mach}}$ is the per-hue angular shift predicted by the Machado model. That model is defined on the Smith and Pokorny cone fundamentals, and we evaluated it on the Stockman and Sharpe fundamentals instead (Stockman & Sharpe, 2000). The gain g parameterizes cortical compensation. At $g = 1$ the cortical stage leaves the retinal distortion unchanged, and at $g = 2$ it fully cancels the retinal shift. At $g > 2$ cortical processing reverses the shift and moves the perceived hue past the undistorted color. The confusion axis is the color-space direction along which CVD observers confound hues with one another. Since $\delta\theta_{\mathrm{Mach}}$ lies entirely along that axis, R+C carries a single free parameter and displaces colors along one fixed direction. $\Delta\lambda$ was not fitted. Each participant was evaluated at three published cone-shift anchors for their subtype, 6.0, 6.5, and 8.0 nm for deutan and 1.5, 3.0, and 10.0 nm for protan, and R+C results are reported at the anchor with the lowest held-out loss.

Candidate Model 2: two-component cortical (2-comp). The second model parameterizes the distortion as a sum of two cosine components, one on the S-cone axis and one on each subtype's confusion axis:

$$\delta\theta(\theta;\, \beta_s, \beta_c) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\mathrm{conf}}) \tag{3}$$

where θ_{conf} is the hue angle assigned to each subtype's confusion direction, 16° for protan and 150° for deutan. Both angles are fixed before fitting. They are defined in a cone-opponent convention and applied to the nominal $L^*a^*b^*$ hue angle of the stimuli, which is an approximation we adopt for the whole model. β_s parameterizes distortion along the S-cone axis ($\theta = 90^\circ$) and β_c along the individual confusion axis. Each component has a fixed direction and a free amplitude, so the model spans distortions in the plane defined by those two axes. Both signs of β_c are unconstrained. We constrain $\beta_s \geq 0$ as a weak directional prior that fixes only the sign, not the magnitude, of the S-cone-axis term, consistent with the perceptual hue circle of protan and deutan observers being indented at yellow and purple-blue, the two ends of the S-cone axis Ohkoba et al., 2022. Model adequacy was assessed by the procedure in §2.12; results are reported in §3.6.

2.11 Inverse fitting

Per-subject distortion parameters (β_s, β_c) were estimated by finding the distortion that, when applied to HC-derived neural representations, best reproduces the observed CVD responses. Three loss families served as the fitting objective. A specific instance of a family — for example, the aggregate JND ratio or the ΔRDM loss at a single ROI — is a *loss atom*; candidate objectives are combinations of atoms, selected as described in §2.12.

Psychophysical JND ratio (L_γ). A distortion changes the perceived angular separation of a hue pair, and a pair that is compressed becomes harder to discriminate. We modeled the predicted threshold for a pair as $\hat{\gamma} = \bar{\gamma}_{\mathrm{HC}}\,(d_{\mathrm{phys}}/d_{\mathrm{perc}})$, where d_{phys} is the separation of the two hues on the display, d_{perc} is their separation after the candidate distortion of Eq. 3, and $\bar{\gamma}_{\mathrm{HC}}$ is the HC mean threshold for that pair. The loss is the squared difference between the predicted and the observed CVD threshold, standardized by the HC standard deviation s_{HC} for that pair and summed over the pairs entering the atom:

$$L_\gamma = \sum_p \left(\frac{\hat{\gamma}_p - \gamma_{\mathrm{CVD},p}}{s_{\mathrm{HC},p}} \right)^2 \tag{4}$$

The atoms available to the fitting were single-pair terms and one aggregate term. For the deutan participant the single-pair candidates were orange–yellow and yellow–green, which bracket that subtype's confusion axis, and yellow–purple, which lies on the S-cone axis. For the protan participant the single-pair candidate was green–blue, which brackets the protan confusion axis. The aggregate term γ_{all} sums over all eight tested pairs. Which atom entered each participant's objective was decided by the procedure in §2.12. The reported ratio $\gamma = \gamma_{\mathrm{CVD}}/\bar{\gamma}_{\mathrm{HC}}$ (§2.3) places the same thresholds on a common scale and serves reporting rather than fitting.

Per-pair RDM difference (ΔRDM). Procrustes disparity summarizes the distance between two geometries as a single value after optimal rotation. It reports how far a participant's geometry lies from the HC reference, and the optimal rotation removes the information about which colors are displaced. Inverting the distortion requires that information. We therefore expressed the same comparison as a difference of representational dissimilarity matrices, which retains the displacement of every color pair. For each participant and ROI the RDM was the 8×8 matrix of pairwise correlation distances $(1 - \text{Pearson}\, r)$ among the reduced color patterns of that ROI, and the per-pair difference was $\Delta\mathrm{RDM} = \mathrm{RDM}_{\mathrm{CVD}} - \overline{\mathrm{RDM}}_{\mathrm{HC}}$, in which $\overline{\mathrm{RDM}}_{\mathrm{HC}}$ is the mean RDM over the seven HC participants.

Representational ΔRDM cosine (L_{RDM}). For a candidate distortion, a simulated difference $\Delta\text{RDM}_{\mathrm{sim}}$ is obtained by applying $\delta\theta$ to the HC mean RDM and comparing it to the observed ΔRDM. Each distorted hue is assigned to the nearest of the eight 45° bins, so $\Delta\text{RDM}_{\mathrm{sim}}$ is built by reindexing the entries of the HC mean RDM. A candidate distortion therefore acts as a permutation of the eight hues, and distortions that map to the same permutation are indistinguishable to this term. The loss is the cosine dissimilarity between the observed and simulated 28-element upper-triangle vectors:

$$L_{\mathrm{RDM}} = 1 - \cos\big(\Delta\text{RDM}_{\mathrm{sim}},\, \Delta\text{RDM}_{\mathrm{obs}}\big) \tag{5}$$

Because the loss is a cosine, the magnitude of ΔRDM does not enter.

The loss computes ΔRDM in a PCA-reduced space ($K = 6$). Across the 28 pairwise entries this estimate agreed with the SRM-aligned estimate used for disparity at V1, V2 and hV4 ($r = 0.77$–0.89), and less closely at V3 ($r = 0.39$–0.58), so the reduction does not determine the pattern the loss fits.

hV4 LOCO voxel-prediction (L_{LOCO}). This atom scores how well a candidate distortion accounts for the CVD participant's own held-out responses at hV4. Weights were fitted within that participant. For each held-out color c, ridge weights W_c were estimated from the remaining seven colors across all six runs, with the penalty chosen by generalized cross-validation as in §2.7. The candidate distortion relabels each stimulus by its perceived hue, so the held-out response is predicted as $\hat{Y}_c = \mathbf{c}(\theta_c + \delta\theta_c)^\top W_c$. The fit for that color is the Pearson correlation ρ_c between $\hat{Y}_c$ and the participant's run-averaged pattern at color c. The loss averages the complement over the eight colors:

$$L_{\mathrm{LOCO}} = \frac{1}{8}\sum_{c=1}^{8}\big(1 - \rho_c\big) \tag{6}$$

Fitting the weights within participant keeps the prediction in that participant's voxel space, since the number of voxels differs across participants and HC-trained weights do not transfer. This atom was available to every candidate objective, and the selection procedure of §2.12 decided whether it entered.

Composite loss. The atoms differ in scale, since L_γ is built from threshold ratios while L_{RDM} and L_{LOCO} are bounded dissimilarities. Each atom was therefore evaluated over the full parameter grid and standardized to zero mean and unit standard deviation across that grid. The composite loss is the sum of the standardized atoms divided by $\sqrt{n_a}$, where n_a is the number of atoms. Standardization sets the relative weight of the atoms and makes any constant rescaling of an individual atom immaterial.

Grid search. Parameters were searched by exhaustive grid at 2° resolution, with β_s over $[0^\circ, 50^\circ]$ (26 values) and β_c over $[-50^\circ, 50^\circ]$ (51 values), giving $1{,}326$ cells. For R+C (Eq. 2), g was searched over $[0, 3]$ in steps of 0.05 (61 values). Per-subject loss combinations and parameter selection are described in §2.12.

2.12 Parameter selection

Per-subject loss combinations and model class were selected by a three-gate procedure. Two resampling schemes over the seven HC references supplied the inputs, a 5-train/2-test resample repeated $N = 300$ times and a strict 7-fold leave-one-HC-out. Both used the same loss atoms.

Gate 1: Separation precondition. Admission required the CVD loss value to differ from the HC leave-one-out distribution by a Cohen's d of magnitude 0.5 or greater, in either direction. Atoms below that threshold were excluded from combination.

Gate 2: Boundary saturation. Among admitted combinations, those in which at least half of the resample solutions saturated the grid boundary were rejected, since boundary saturation indicates model misspecification. Combinations were also rejected when the recovered parameters collapsed across resamples, defined as an interquartile range above 50° or a sign reversal between the training and test argmin exceeding 5°.

Gate 3: Held-out test-loss. Surviving combinations were ranked by the median composite test-loss $\overline{L}_{\text{test}}$ across the $N = 300$ resamples. For each resample the composite loss was evaluated at the training argmin on the two held-out HC and z-normalized relative to the training grid, so a lower $\overline{L}_{\text{test}}$ indicates generalization to unseen HC references. Ties were broken by the test-loss interquartile range, which measures the stability of that generalization estimate. These two quantities were the only ranking criteria.

Further quantities were recorded for each candidate outside the ranking. The held-out neural loss $\overline{L}_{\text{RDM}}^{\text{LOO}}$ from the strict 7-fold leave-one-out evaluates a candidate's neural atoms with the psychophysical term excluded. It therefore isolates what the neural terms alone contribute to the fit. Parameter stability was summarized by the interquartile range of the recovered $(\hat{\beta}_s, \hat{\beta}_c)$, by the fraction of resamples recovering the modal 45° bin, and by the strict 7-fold parameter range. The HC false-positive rate is likewise reported descriptively in §3.8.

2.13 Identifiability and recovery

Four pre-specified checks were run on each production candidate before any parameter was reported.

Parameter recovery. This test establishes whether a known distortion can be recovered at the present sample size and noise level. Synthetic CVD responses were generated at a known ground truth by passing that distortion through each HC participant's encoder. Noise matched that participant's residual structure, using the top 20 principal components of its spatial covariance and an AR(1) correlation of 0.3 across runs. The grid fit was re-run on each synthetic dataset, giving $n = 140$ refits per candidate from seven HC encoders and twenty noise realizations. Recovery was summarized by f_{10°, the fraction of refits within 10° of ground truth on both axes, against a pass criterion of $f_{10^\circ} \geq 0.5$ with $|\text{bias}| < 10^\circ$. The refit repeats the grid search rather than the full selection procedure, so it measures recoverability at a fixed loss combination.

Origin-recovery null. This test records the spread the procedure returns when no distortion is present, which sets its per-axis uncertainty floor. The same synthesis was run with ground truth at $(0^\circ, 0^\circ)$ and each HC participant's real JND. It returns no p-value and was excluded from the test pool.

HC pseudo-CVD specificity. This test asks whether a HC participant placed in the CVD position yields a comparable estimate. Each HC participant's real amplitudes were substituted into the CVD slot in turn, giving a seven-member null and a one-sided rank-distance percentile.

Color-label permutation. This test asks whether the estimate depends on which color evoked which response. The eight color labels of the real CVD amplitudes were permuted $N = 1000$ times, with the same permutation applied across ROIs and the HC pool left unchanged, giving a one-sided p_{perm}.

The three test-bearing checks across the two production candidates gave six tests, corrected together under the Benjamini–Hochberg false-discovery rate ($\alpha = 0.05$). Outcomes are reported in §3.8.

2.14 Stimulus-space filter

The correction filter for each participant is the pre-image of the fitted transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the eight displayed hues θ_k we solved $T(\tilde{\theta}_k) = \theta_k$ for the input angle $\tilde{\theta}_k$, which gives a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. Under the model, a CVD observer viewing $\tilde{\theta}_k$ then obtains the cortical hue representation that a healthy control observer obtains when viewing θ_k.

Each pre-image was located by Brent root-finding on the residual $T(\theta) - \theta_k$, searched within $\pm 60°$ of the target and refined to a bracket tolerance of 10^{-9} degrees. The transform is not guaranteed to be invertible for arbitrary parameters, so every solution was verified by forward-substitution. All eight hues resolved to within 10^{-3} degrees of their target for both deployed parameter sets, and the procedure rejects a parameter set that leaves any hue unresolved.

2.15 Filter evaluation

Both CVD participants completed a second scanning session that compared the individualized filter and a deployed macOS accessibility filter against the unfiltered Session-1 baseline. The individualized filter was the per-subject 2-component stimulus-space pre-image, with parameters frozen from the main analysis and therefore out-of-sample. Acquisition, comparator, run-count, and single-case-inference details are given in Supplementary §4.5.

On the second-session data we recomputed, within each condition, the primary Session-1 interpolation metric (LOCO adjacent accuracy, §2.8), LORO classification, SRM Procrustes disparity into the HC shared space, and the representational dissimilarity matrices. As a secondary encoding index we also computed the forward-tuning correlation, the Pearson r between the predicted and the observed voxel pattern at each held-out hue, averaged over the eight hues. It indexes the encoding fit rather than the decoding accuracy that carries the comparison, and is reported as corroboration only.

Each condition's neural index was reported as a standardized single-case effect size (Cohen's d against the HC distribution) rather than an inferential p, because the grand-mean permutation null is biased for this design. In both CVD participants, the JND and 8AFC tasks (§2.3) were repeated under each filter, each compared to the unfiltered Session-1 baseline and to the HC distribution. This interleaved, cross-session psychophysical comparison is descriptive.

2.16 Reproducibility

All analyses were conducted in Python 3.10 with numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0 (Pedregosa et al., 2011), and BrainIAK 0.11. Random seeds were fixed for every permutation test and bootstrap. Most analyses used seed 42, while the parameter-recovery synthesis used 31337 and the color-label permutation used 27182, and each script records its own value. Analysis code and preprocessing scripts are openly available at https://github.com/Transconnectome/colorBlind_analysis. The neuroimaging data are held under the access conditions imposed by the approved protocol, which are stated in full in the Data and Code Availability section.

3 Results

3.1 All eight colors remain decodable across ROIs in both CVD cases

Both CVD participants classified all eight hues above the 0.125 chance level at every ROI (Figure 4A). This satisfies the precondition for a corrective stimulus-space filter, which requires the target hues to remain decodable from the visual cortex of the individual being corrected. Accuracies come from leave-one-run-out (LORO) cross-validation of the forward encoding model on the Procrustes-aligned amplitudes. The lowest CVD value was 0.375 at hV4, three times the chance level. Repeating the readout in the SRM-aligned space reproduces the pattern (Supplementary §S11).

An encoder built from the healthy controls also transferred to the CVD cortex. Applied to the CVD participants, it decoded their held-out runs at 0.432 on average over the eight participant-by-ROI cells, far above the 0.125 chance level ($t(7) = 6.51$, $p < 0.001$). The same encoder reached 0.526 over the 28 cells of the held-out controls (Mann–Whitney $U = 163.5$, $p = 0.052$, rank-biserial $r = 0.46$). Transfer is therefore attenuated rather than abolished, which locates the CVD responses within the channel-to-voxel code that the control encoder describes. Single-case comparisons of each participant's own accuracy against the controls appear in Supplementary §S21, where none of those eight cells reaches significance. Both the within-subject readout and the control-derived encoder therefore recover the eight stimuli from CVD cortex, consistent with prior work showing that above-threshold color differences in anomalous trichromacy are better preserved than threshold sensitivity predicts (Boehm et al., 2014).

3.2 Hue interpolation is reduced at hV4 in both CVD cases

In the healthy controls, hue interpolation succeeded at hV4 alone (adjacent accuracy 0.456 ± 0.039, mean $\pm$ SEM over $n = 7$, permutation $p = 0.011$). Interpolation was measured by leave-one-color-out (LOCO) decoding of the forward encoding model. Its readout selects among 360 integer hues and scores a prediction correct within $45°$, so the analytic chance level is 0.25 rather than the 0.125 of the eight-way classification above (Section 2.8). Each ROI was also tested against a null of $1{,}000$ per-subject color-label permutations. That null lay near 0.35, above the analytic level, because shuffling the labels leaves the covariance among voxels intact. All four ROIs exceeded 0.25, and V1 to V3 stayed within their own permutation nulls (V1 $p = 0.164$, V2 $p = 0.424$, V3 $p = 0.586$). This replicates Brouwer and Heeger (2009) and fixes hV4 as the ROI where a CVD reduction in interpolation is interpretable.

Both CVD participants fell below the control distribution at hV4, significantly so in the protan participant (Crawford & Howell $t = -3.04$, $p = 0.012$, $d_{cc} = -3.25$) and with a large effect in the deutan participant ($t = -1.89$, $p = 0.054$, $d_{cc} = -2.02$). Adjacent accuracy was 0.13 in the protan participant and 0.25 in the deutan participant, against a control mean of 0.46 (Figure 4B). The deutan p-value reflects the power of a single-case test against $n = 7$ controls.

The reduction concentrated on the S-cone intermediate hues. Both participants gave zero adjacent accuracy at hV4 for blue, purple, and magenta (Figure 4C). Per-hue single-case tests reached $p \geq 0.051$ at every hue (Crawford & Howell, one-tailed, uncorrected and exploratory), with blue the largest deviation in both participants ($d_{cc} = -2.06$). The per-hue accuracies form each participant's hue vulnerability profile $\mathbf{v} \in [0, 1]^8$.

3.3 Geometric deviation localizes to a distinct ROI in each CVD case

Elevated Procrustes disparity localized to a different ROI in each CVD participant, V1 in the protan participant and V2 in the deutan participant (Figure 5B). Procrustes disparity measures how far a participant's whole hue geometry lies from the healthy-control reference in the shared SRM-aligned space.

The protan participant showed elevated disparity specifically at V1, under both the common HC space ($p = 0.007$) and the symmetric LOSO reference ($p = 0.045$). Disparity at V2, V3, and hV4 stayed within the control range ($p \geq 0.150$).

The deutan participant showed elevated disparity at V2 in the common HC space ($p = 0.040$), which did not survive the symmetric LOSO reference ($p = 0.116$, $d_{cc} = 1.42$). V3 in the same participant reached $p = 0.052$ ($d_{cc} = 2.05$), so V2 and V3 are not separable here. The LOSO reference widened the confidence interval without changing the direction or magnitude of the V2 effect. Distinguishing an attenuated effect from an absent one requires a larger cohort. Full test statistics and effect sizes (t, d_{cc}) for both estimators are reported in table S5.

The two CVD participants differ in where the deviation lies and in how strong it is. Because each deviation is participant-specific, a single family-level correction would match only one of them. In each participant the continuous arrangement of hues is displaced in a direction and magnitude specific to that

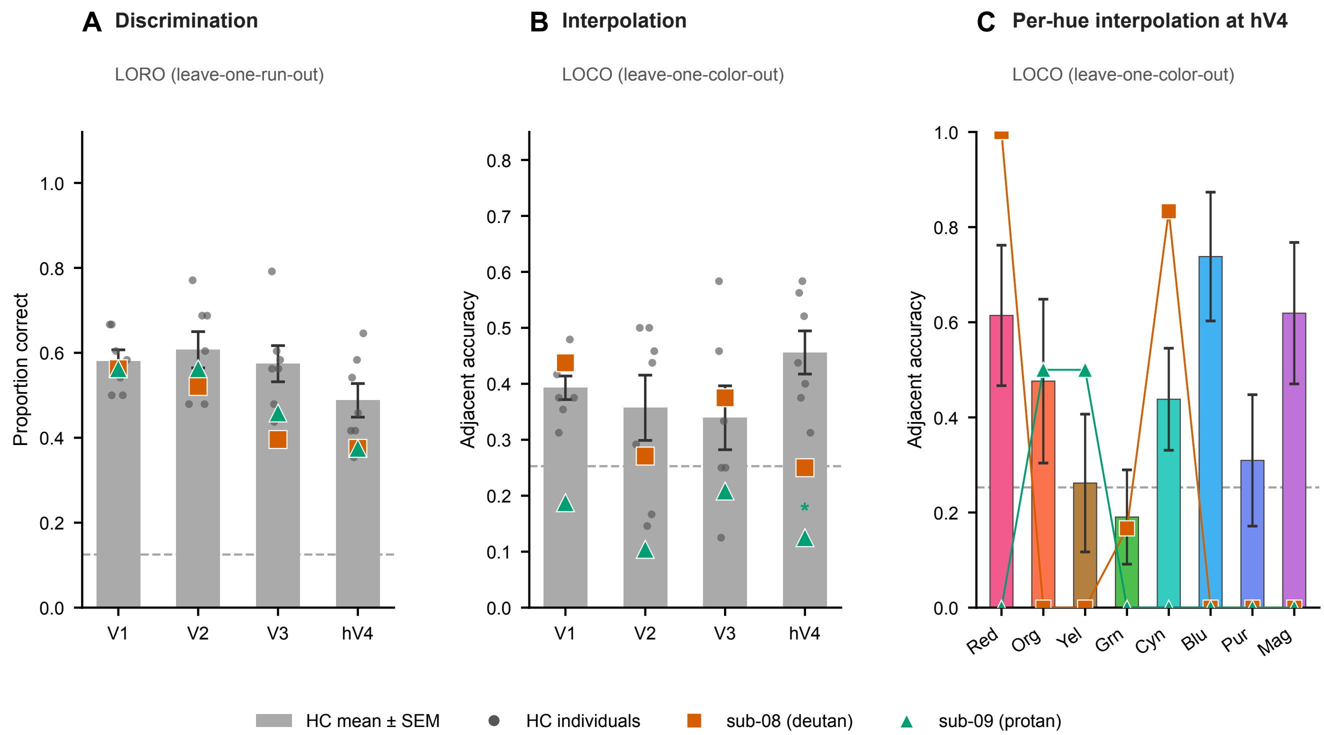

Figure 4: **Color classification and hue interpolation across V1–hV4 in the two CVD participants and healthy controls.** All panels are computed on the Procrustes-aligned amplitudes with $n = 7$ healthy controls at every ROI. Gray bars give the control mean $\pm$ SEM, the orange square the deutan participant, and the teal triangle the protan participant. **(A)** Eight-way classification accuracy from leave-one-run-out cross-validation of the forward encoding model. The dashed line marks the chance level of $1/8$. Single-case Crawford–Howell comparisons are two-tailed here, since the hypothesis is preservation. **(B)** Adjacent accuracy from leave-one-color-out cross-validation, the proportion of predictions falling within $45°$ of the target hue. The dashed line marks the analytic chance level of 0.25. Accuracies are shown for all four ROIs. Asterisks give one-tailed single-case Crawford–Howell comparisons and appear only at ROIs whose control mean exceeds the color-label permutation null. **(C)** Per-hue adjacent accuracy at hV4. Per-hue comparisons are one-tailed, uncorrected, and exploratory across the eight hues.

individual, which a personalized correction must therefore offset.

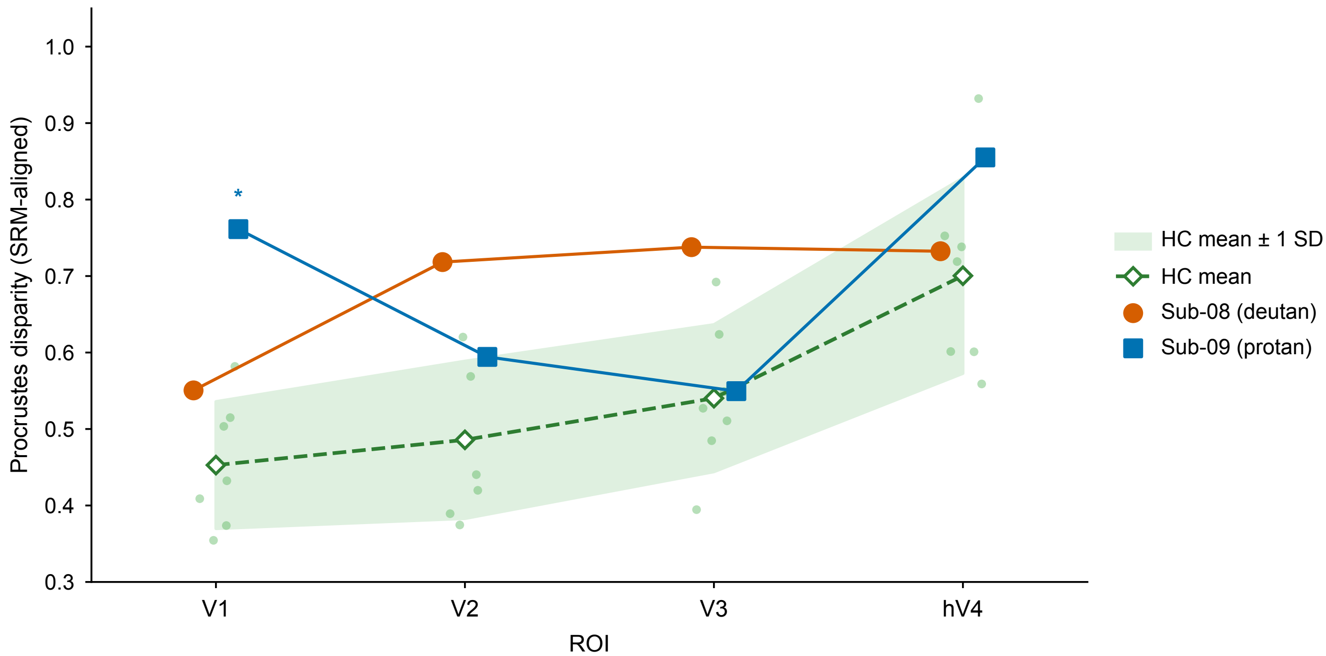

Figure 5: **Procrustes disparity to the healthy-control reference.** Disparity per participant and ROI, indexing the distance of the whole hue geometry from that reference. The gray band gives the HC group mean ± 1 SD ($n = 7$) and the dots give the individual HC leave-one-out values. Single-case comparisons used the Crawford–Howell modified t against the HC leave-one-out distribution, one-tailed. Asterisks mark $p < 0.05$ under the symmetric LOSO reference, which is the inferential estimate (§2.9). Test statistics under both references are given in table S5.

3.4 Hue-discrimination thresholds are elevated on each participant's own confusion axis

Each CVD participant's threshold elevation fell on the confusion axis of their own subtype. In the deutan participant the two pairs bracketing the deutan axis were elevated, orange–yellow ($\gamma = 3.02$, $z = +4.15$) and yellow–green ($\gamma = 3.10$, $z = +4.32$), as was the S-cone pair yellow–purple ($\gamma = 2.87$, $z = +6.70$). In the protan participant the elevation was confined to green–blue, which brackets the protan axis ($\gamma = 1.95$, $z = +2.36$). Here γ is the ratio of the participant's threshold to the healthy-control mean and z is the deviation in control standard deviations ($n = 7$).

The remaining pairs lay within the control range in both participants ($|z| \leq 1.23$). This includes red–cyan, the $180°$ control pair, where both participants discriminated more finely than the controls ($\gamma = 0.45$), which places the elevations away from a ceiling effect. Averaged over the eight pairs, mean

$|z|$ reached 2.24 in the deutan and 0.90 in the protan participant. Values for all eight pairs appear in Supplementary §S15.

These ratios supply the psychophysical atoms available to the fitting (§2.10) and stand alongside the neural geometry as the second input to the correction.

3.5 The retinal-plus-gain model is structurally insufficient

The R+C model is structurally insufficient for both CVD participants, though the supporting evidence differs by case. R+C fixes the retinal cone shift $\Delta\lambda$ at a published value and fits only the cortical gain g. Each participant was therefore fitted at three $\Delta\lambda$ anchors, and R+C is reported at the anchor with the lowest held-out loss (Section 2.10). In the deutan participant every resample solution saturated the grid boundary at $g = 3.0$, which exceeds the 50% boundary-saturation gate (Section 2.12) and rejects the model as misspecified (Wilson & Collins, 2019). In the protan participant 41% of solutions saturated, which leaves that gate untripped, but the fit still settled at $g = 2.95$. The 2-component model also generalized better there ($\overline{L}_{\text{test}} = -1.54$ against -0.86, Section 3.6).

A gain $g > 2$ implies that cortical processing reverses the direction of the retinal shift past the undistorted hue (Section 2.10). Both participants scored below the Ishihara criterion, the deutan participant identifying 5 of 14 plates and the protan participant 7 of 14, which confirms residual CVD under standard viewing. A fitted $g > 2$ therefore contradicts the measured status of both participants, since the model claims overcompensation where they show impairment. The saturated boundary indicates misspecification rather than a parameter estimate.

The R+C family also carries a degree-of-freedom restriction. The function $\delta\theta = (2-g)\cdot\delta\theta_{\text{Mach}}$ displaces hues along the fixed confusion axis alone, for every value of g. It therefore omits displacement away from that axis, including along the S-cone axis of the 2-component model. R+C is excluded as a filter candidate.

3.6 A common cortical model fits both CVD cases

The 2-component cortical model fit both participants with the same functional form, and the deutan and protan cases differ only in $(\hat{\beta}_s, \hat{\beta}_c)$. Both winning combinations paired a JND atom with a ΔRDM atom, and the LOCO family entered neither.

For the deutan participant ($\theta_{\text{conf}} = 150°$), the selected loss combination $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) =$

$(6°, -42°)$. The held-out test-loss was $\overline{L}_{\text{test}} = -2.36$ (IQR $= 2.15$), against -1.14 for the competing β_s-dominant candidate $(38°, -10°)$. The HC-resample parameter IQR was $(8°, 2°)$. Across the strict 7-fold leave-one-out refit $\hat{\beta}_c$ ranged from $-46°$ to $-38°$, so its sign held on every held-out reference set.

For the protan participant ($\theta_{\text{conf}} = 16°$), the selected loss combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) = (2°, +24°)$. The held-out test-loss was $\overline{L}_{\text{test}} = -1.54$ (IQR $= 1.42$). The runner-up loss combination replaced γ_{all} with γ_{GB} and reached -1.52 at the same estimate $(2°, +24°)$, so the fitted parameters do not depend on which psychophysical atom enters. Both combinations improved on the retinal-plus-cortical class scored on the same composite ($\overline{L}_{\text{test}} = -0.86$). The HC-resample parameter IQR was $(0°, 0°)$, and 263 of the 300 resamples returned the identical cell $(2°, +24°)$, with the remainder at $(32°, 0°)$. The strict 7-fold leave-one-out refit returned $(2°, +24°)$ on all seven folds. This stability holds for the PCA-basis loss. The argmin shifts under the alternative SRM-basis reduction, so the recovered position is metric-dependent (Supplementary §4.5).

The separation precondition left the RDM atom at all four ROIs in the deutan participant and at V1 alone in the protan participant (four entries and one on the ROI axis of the combination grid). The RDM atom entered the grid only at ROIs where it separated the participant from the control distribution (Section 2.12). The psychophysical axis was enumerated in full. For the RDM atom that criterion gave $d = 2.31$, 1.94, 0.86, and 2.19 across V1 to hV4 in the deutan participant, against 0.81, -0.23, -0.48, and -0.24 in the protan participant. Within the deutan grid, $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ ranked first among the 25 combinations passing the boundary-saturation gate. The fitted parameters are insensitive to that choice. In the deutan participant V2 and V3 both return $(6°, -42°)$, and dropping the RDM atom returns $(16°, -44°)$. Substituting V2 or V1+V4 for V1 in the protan participant also returns $(2°, +24°)$.

Both fits generalized to held-out control data. On each of the seven leave-one-out folds the CVD representational geometry was predicted more closely by the fitted distortion than by no distortion at all. On held-out loss the selected cell ranked in the top 5% of the $1{,}326$-cell parameter grid for the deutan participant and the top 8% for the protan. Held-out losses for both fits are listed in table S11.

3.7 The neural term relocates the protan fit and sharpens the deutan fit

The neural term moved the argmin in the protan participant, sharpened it in the deutan participant, and narrowed the resample spread in both.

In the protan participant the RDM atom moves the estimate from the S-cone-dominant $(26°, +4°)$ of the psychophysical fit to the confusion-axis-dominant $(2°, +24°)$. At that estimate the psychophysical term matches the no-distortion control ($\Delta L = +0.01$, below it on 3 of 7 folds), whereas the RDM term falls 0.47 below it. The confusion-axis component of this participant is carried by the neural data.

In the deutan participant the RDM atom adds precision rather than direction. The psychophysical atom alone already settles at $(16°, -44°)$, against $(6°, -42°)$ for the combined fit. Adding the RDM atom lowers the boundary-saturation rate from 23% to 9.3%, and a neural-only fit reaches $(4°, -26°)$ on the same side of the confusion axis.

The resample spread narrowed in both participants. In the deutan participant the HC-resample IQR fell from $(18°, 6°)$ to $(8°, 2°)$ under the PCA basis and to $(10°, 4°)$ under the SRM basis. In the protan participant it fell from $(6°, 4°)$ to $(0°, 0°)$ under the PCA basis and to $(0°, 2°)$ under the SRM basis, where the argmin sits at $(32°, +0°)$ rather than at the PCA solution.

3.8 The mechanism class is recoverable while the per-axis magnitudes are bounded rather than estimated

The deutan confusion-axis sign is recoverable, returning $\hat{\beta}_c < 0$ in all 300 resamples and all seven leave-one-out folds under both the PCA-basis and the SRM-basis loss. The protan sign is basis-dependent. Under the PCA-basis loss the fit returns $(2°, +24°)$ in 263 of 300 resamples and on all seven folds, whereas under the SRM-basis loss it returns $(32°, 0°)$ in 171 of 300, with $\hat{\beta}_c$ positive in 17% of resamples and negative in 26% (Supplementary §S18). Each participant's selected fit is the global minimum on the full data (Figure 6). The $(32°, 0°)$ solution is the secondary basin of the protan PCA landscape, so the two reductions disagree about which basin is the minimum.

Only the deutan confusion-axis amplitude exceeds the error of the recovery procedure. Refitting synthetic data with known parameters left an effective uncertainty of $22°$ on β_s and $26°$ on β_c for the deutan candidate, and $16°$ and $24°$ for the protan candidate. The deutan $|\hat{\beta}_c| = 42°$ exceeds its uncertainty and was recovered to $4.7°$ on average, whereas the protan $|\hat{\beta}_c| = 24°$ equals its uncertainty and was recovered to $26.4°$. The fitted $|\hat{\beta}_s|$ of $6°$ and $2°$ falls well below the uncertainty on that axis, so the S-cone amplitude cannot be distinguished from zero at this resolution. Recovery within $10°$ succeeded on 26% of draws for the deutan candidate and 14% for the protan, and six pre-specified checks across the two candidates returned no significant result after Benjamini–Hochberg correction (Supplementary §S18).

Within the deutan fit the two axes are constrained to different degrees. Across the 300 resamples $\hat{\beta}_c$ concentrates at $-42°$ and $-44°$ $(202$ of $300)$ and stays within $[-48°, -36°]$, whereas $\hat{\beta}_s$ distributes almost uniformly from $0°$ to $14°$.

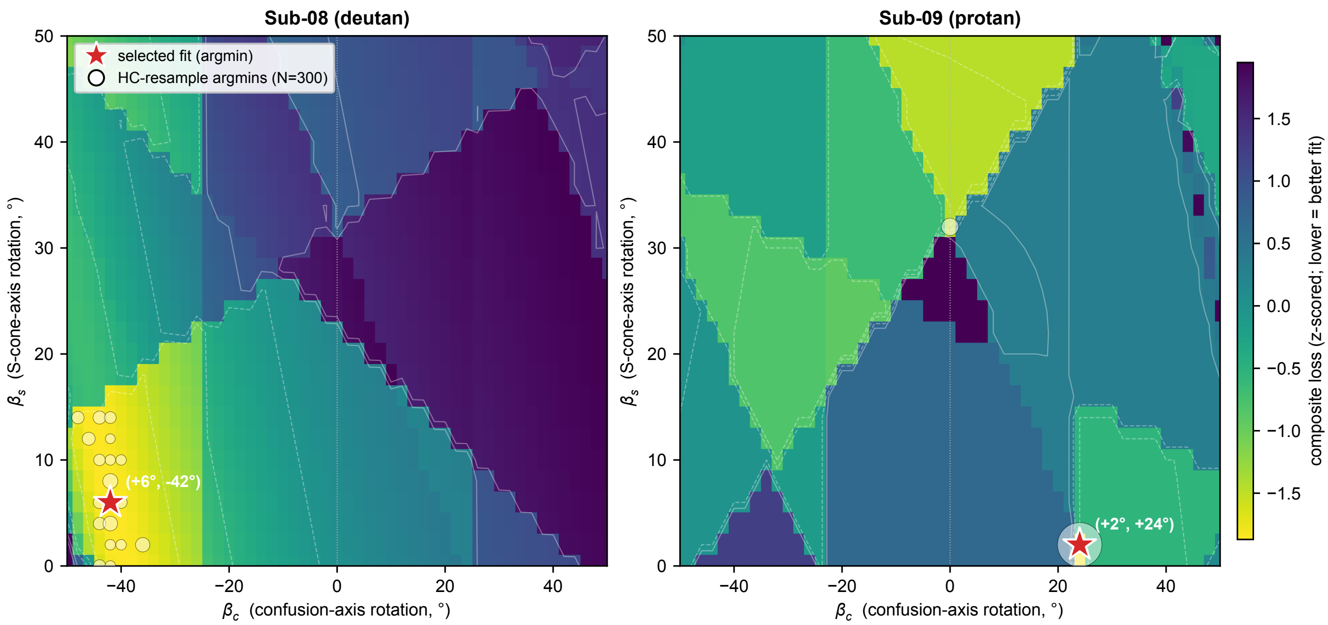

Figure 6: **Per-subject loss landscape and parameter uncertainty.** The z-scored composite loss over the (β_s, β_c) grid, reconstructed on the full 7-HC pool, with lower values in yellow. The loss combination is $\gamma_{\rm OY} + L^{(V2)}_{\rm RDM}$ for the deutan participant and $\gamma_{\rm all} + L^{(V1)}_{\rm RDM}$ for the protan participant. The red star marks the argmin of that surface, at $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$ for the deutan participant and $(2^\circ, +24^\circ)$ for the protan participant. It is a descriptive low-dimensional embedding of the distortion rather than a physiological point estimate (Section 3.8). White points give the per-resample argmins over $N = 300$ HC 5-train/2-test resamples.

3.9 Per-subject stimulus-space filter

The two filters differ in mean amplitude by 10° and in direction at two of the eight hues. Each is the exact numerical pre-image of that participant's fitted 2-component transform (Section 2.14).

For the deutan participant ($\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -42^\circ$, $\theta_{\rm conf} = 150^\circ$) the per-hue correction ranges from 9.1° to 37.9° in magnitude, with a mean of 26.3° (Figure 7). For the protan participant ($\hat{\beta}_s = 2^\circ$, $\hat{\beta}_c = +24^\circ$, $\theta_{\rm conf} = 16^\circ$) it ranges from 6.1° to 24.6°, with a mean of 16.2°.

The dominant confusion-axis term carries opposite signs in the two participants, -42° against $+24^\circ$, and their confusion axes differ as well, at 150° against 16°. The two effects partly cancel, so the resulting corrections share a direction at six of the eight hues and oppose each other at yellow and purple.

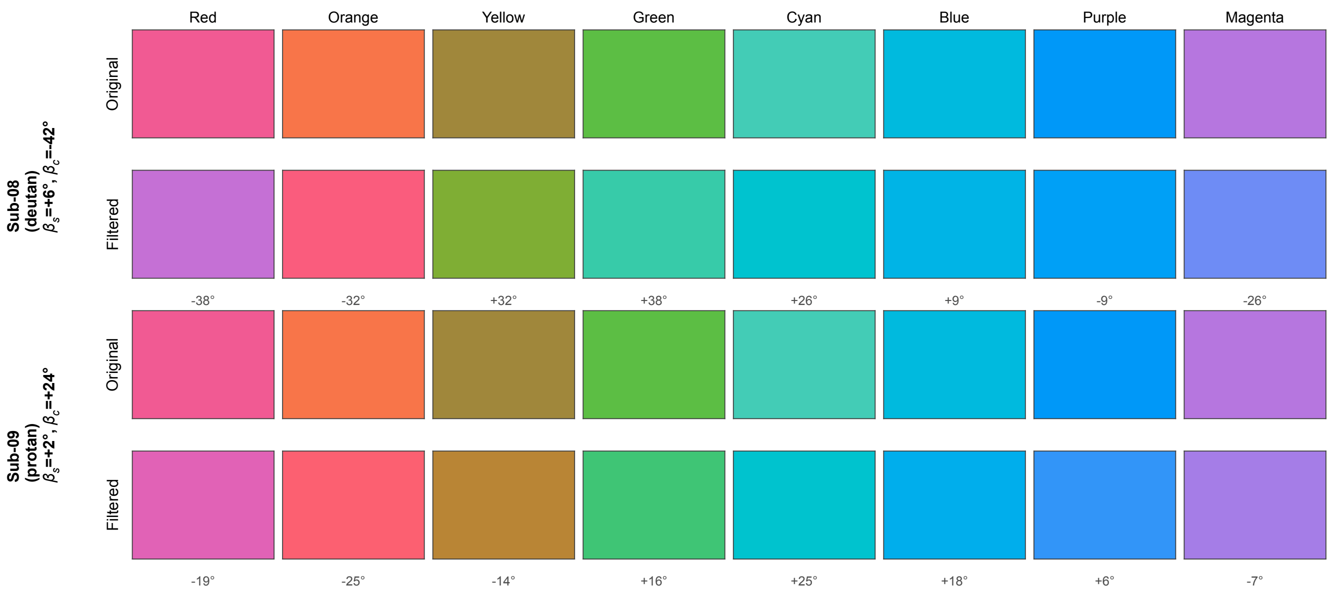

Figure 7: **Per-subject stimulus-space correction filter.** For each CVD participant, the displayed stimulus (*Original*) and the corrected stimulus (*Filtered*) are shown for the eight scanner hues, which run across the columns. Top block: the deutan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (6°, -42°)$. Bottom block: the protan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (2°, +24°)$. The filtered stimulus is the exact numerical pre-image of the fitted 2-component transform. The per-hue correction $\delta\theta_{\text{filter}} = \theta_{\text{pre}} - \theta$ is annotated beneath each *Filtered* swatch (8/8 pre-images exact, residual $< 0.001°$ for both subjects). Rendering coordinate space: STIM_LAB CIELab (project convention). Neural validation of both filters in a second session is shown in fig. 8. The psychophysical outcomes are reported in Section 3.10.

3.10 Psychophysical and neural filter evaluation

Filter evaluation. Both CVD participants completed a second session under each filter, repeating the psychophysical tasks and the Session-1 neural readouts (Supplementary §S19).

Psychophysics. The individualized filter lowered the mean threshold deviation from the controls in the deutan participant and left it unchanged in the protan participant, whereas the deployed filter lowered it in the deutan participant and raised it in the protan one. Session-1 baselines are reported in Section 3.4.

In the deutan participant both filters removed the baseline deficit. The three elevated pairs fell from $+4.2$, $+4.3$, and $+6.7$ control SDs to within ± 1.8 under both filters, and the mean $|z|$ fell from 2.24 to 0.85 under the deployed filter and 0.78 under the individualized one. 8AFC identification rose from 0.81 (95% CI $[0.70, 0.89]$) to 0.97 ($[0.89, 0.99]$) under both (Wilson score interval, $n = 64$). The two filters were indistinguishable on thresholds (Wilcoxon $p = 0.84$).

In the protan participant the two filters diverged. The deployed filter raised the green–blue threshold from

$+2.4$ to $+5.1$ control SDs ($p = 0.003$) and the cyan–magenta threshold from -0.3 to $+6.2$ ($p = 0.001$), lifting the mean $|z|$ from 0.90 to 1.78. The individualized filter lowered green–blue to $+0.9$ ($p = 0.44$), held every pair within ± 1.5, and left the mean $|z|$ at 0.93. 8AFC identification was at ceiling without a filter (1.00, 95% CI $[0.94, 1.00]$), held at 0.98 ($[0.92, 1.00]$) under the individualized filter, and fell to 0.86 ($[0.75, 0.92]$) under the deployed one.

Only the individualized filter left both participants free of new deviant pairs. Establishing whether it outperforms the deployed filter requires more participants.

Colors remained decodable. All eight hues remained decodable under both filters in both participants. LORO eight-way classification accuracy stayed above the 0.125 chance level at every ROI and in every condition, with the lowest cell at 0.50 against an HC range of 0.71 to 0.77 (Supplementary Table S16).

Interpolation. At hV4 the individualized filter raised interpolation in the deutan participant and lowered it in the protan participant. Adjacent accuracy in the deutan participant went from 0.23 unfiltered to 0.31 under the individualized filter and 0.25 under the deployed one, against an HC mean of 0.46 ± 0.11. Both conditions stayed below the HC level ($d_{cc} = -1.35$ individualized, -1.94 deployed), and the individualized value alone exceeded the 0.25 chance level. In the protan participant the individualized filter gave the lowest of the three conditions at 0.06 ($d_{cc} = -3.70$), against 0.14 unfiltered and 0.19 deployed, and all three fell below chance.

Interpolation is reported at hV4 alone. V1 to V3 are shown in Figure 8D for completeness, but the controls do not interpolate above the color-label permutation null there (Section 3.2), so a filter-induced change in those ROIs carries no interpretation. A secondary forward-tuning encoding index corroborated the deutan improvement at hV4 ($\rho = +0.18$ against HC $+0.21$, deployed $\rho = -0.39$) and did not separate the conditions in the protan participant (Fig. S2).

Geometry. Both filters left the representational geometry short of the HC level, and the direction of the filter effect reversed between the two participants (Fig. 8B,C,E,F). All values below use the run-matched estimates of Supplementary §S19, in which the HC reference and the unfiltered baseline are built from four runs to match the filter conditions. The HC reference is a disparity of 0.44 at V2 and 0.43 at V1, with an RDM self-consistency of 0.59 and 0.66.

In the deutan participant the unfiltered baseline lay closest to HC and both filters moved the geometry further away. V2 SRM disparity went from 0.68 unfiltered to 0.87 deployed and 0.77 individualized. V2

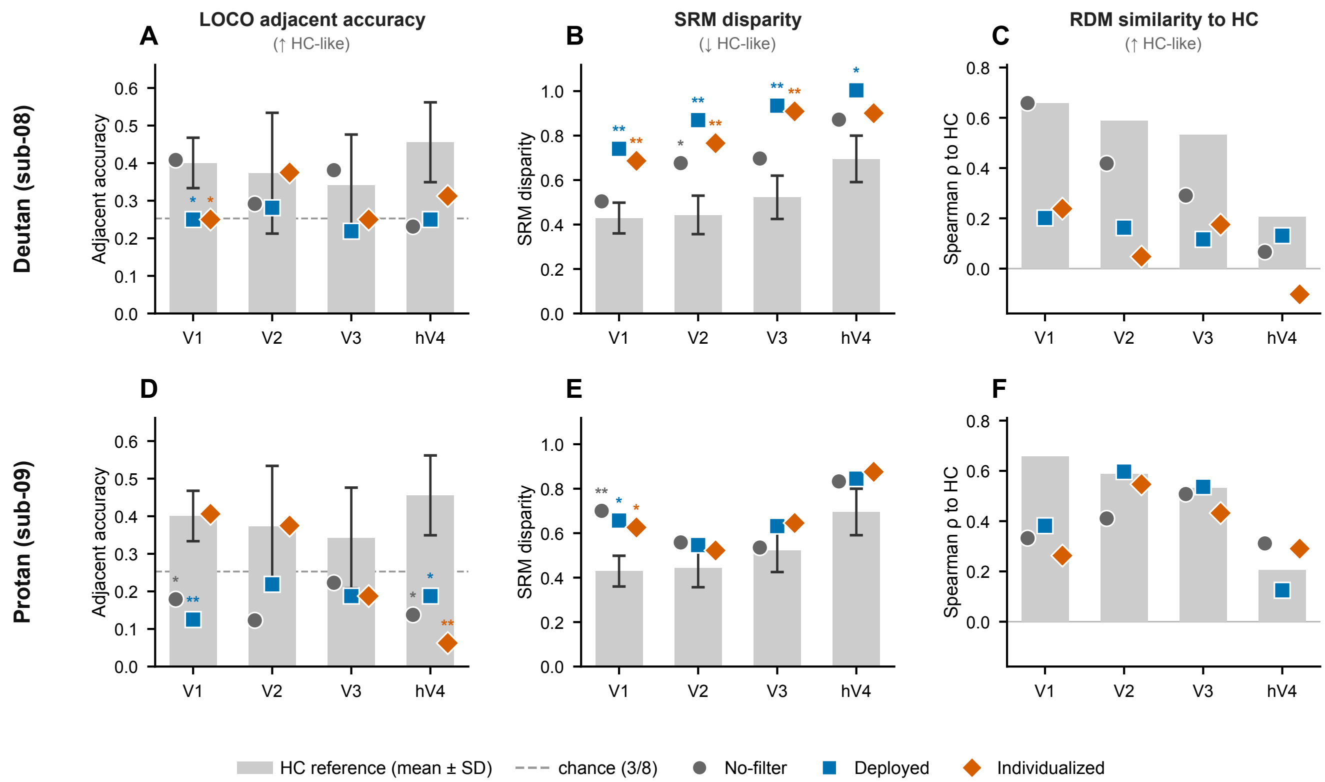

Figure 8: **Hue interpolation and representational geometry under each filter in the second session.** Rows: deutan (sub-08, A–C) and protan (sub-09, D–F). Columns: interpolation and representational geometry across V1–hV4. Markers: no-filter (gray circle, Session-1 baseline), deployed accessibility filter (blue square), individualized filter (orange diamond). Every panel is run-matched, with the HC reference and the unfiltered baseline built from four runs to match the filter conditions (Supplementary §S19). The HC reference is $n = 7$ at V1–V3 and $n = 6$ at hV4, where one control has too few voxels. **(A, D)** LOCO adjacent accuracy, the proportion of predictions within one hue step of the target (↑ HC-like). Gray bars give the HC mean $\pm$ SD and the dashed line the analytic chance level of 0.25. hV4 is the region credited with interpolation (Section 3.2). **(B, E)** SRM disparity into the HC shared space (↓ HC-like). Gray bars give the HC mean $\pm$ SD. **(C, F)** Shared-space RDM similarity to HC (↑ HC-like). The gray bar is the HC leave-one-out self-consistency, a single value with no dispersion, so these panels carry no error bar and no significance marker. Stars in (A, B, D, E) give one-tailed Crawford–Howell single-case deviations from the HC distribution ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$). The forward-tuning encoding index is in Fig. S2.

RDM similarity fell from 0.42 unfiltered to 0.16 deployed and 0.05 individualized.

In the protan participant the unfiltered baseline lay furthest from HC, and the two indices disagreed. V1 SRM disparity went from 0.70 unfiltered to 0.66 deployed and 0.63 individualized, so both filters moved the geometry closer. V1 RDM similarity rose from 0.33 unfiltered to 0.38 under the deployed filter and fell to 0.26 under the individualized one.

The protan gain on disparity was comparable between the two filters. On RDM similarity only the deployed filter improved on the unfiltered baseline. Neither index attributes the geometric recovery to individualization.

4 Discussion

We derived a per-person color-correction filter for each of two CVD participants. Each filter inverts a model of one participant's cortical color representation, fitted to that participant's own neural and psychophysical measurements. That representation carried a structured geometric distortion, which a uniform attenuation of the chromatic signal does not account for. Continuous-hue interpolation fell to chance in both participants, and representational disparity from the healthy-control (HC) geometry exceeded the disparity measured among controls themselves, while all eight hues remained decodable and overall response magnitude stayed within the control range at every early-visual ROI. We modeled the distortion as a hue rotation about two axes of color space, the subtype confusion axis and the S-cone axis. Both axes were fixed before fitting (Methods §2.11). The confusion axis is each subtype's classical confusion-line direction, taken from cone fundamentals. The S-cone axis is the direction in which S-cone excitation varies, and the interpolation deficit concentrated along it. Inverting the fitted model gives a replacement color for any target hue. We applied it at the eight stimulus hues. In a two-person evaluation, the individualized filter removed the deutan participant's baseline threshold deficit and normalized a trending confusion-axis pair in the protan participant, while every other pair stayed within the HC range. The deployed accessibility filter instead rendered two pairs deviant in the protan participant. On cortical hue interpolation the filter's effect varied by participant, with adjacent-hue accuracy increasing in the deutan participant and decreasing in the protan participant. Establishing an advantage over the deployed filter in a larger sample is the immediate next test.

4.1 A geometric distortion of cortical color representation

In both CVD participants the deficit took the form of a structured distortion of cortical color geometry, localized to a different area in each. Two complementary measurements resolve it. The representational dissimilarity matrix (RDM) encodes the pairwise distance structure among stimuli within a cortical region (Kriegeskorte et al., 2008), and it identifies which of those distances depart from the HC geometry. Disparity from HC was elevated at V2 in the deutan participant and at V1 in the protan participant (Section 3.3). Leave-one-color-out decoding identifies where interpolation across the continuous hue circle falls to chance. In the HC cohort, hV4 alone supported above-chance interpolation, replicating Brouwer and Heeger (2009). Both CVD participants fell at or below chance at hV4, most severely at the S-cone intermediate hues blue, purple, and magenta (Section 3.2). This pattern aligns with prior characterizations of CVD as a multidimensional deformation of color space rather than a uniform attenuation (Boehm et al., 2014; Emery et al., 2021; Ohkoba et al., 2022). That the two participants' distortions localized to different areas is consistent with the perceptual case. Among anomalous trichromats, between-observer variability in hue scaling is 3.4 times the within-observer variability (Emery et al., 2021).

4.2 A neurally grounded, individualizable correction filter

Inverting each participant's fitted two-component model assigns every hue a single replacement color (Methods §2.14). That replacement is the correction filter, the displayed hue that the participant's own cortical transformation maps onto the intended color. Across the eight hues the filter applied a mean rotation of $|\delta\theta| = 26.3°$ for the deutan participant and $16.2°$ for the protan participant.

To our knowledge, this is the first color-correction filter derived from an individual's cortical color representation rather than from a retinal model. Its parameters come entirely from that individual's measured cortical color geometry, whereas population-average corrections apply a fixed subtype-average spectral shift. Prior inversions of cortical encoding models set the magnitude of response in a chosen unit or region rather than the arrangement of colors within it (Bashivan et al., 2019; Shinkle & Lescroart, 2025).

The neural term made three distinct contributions to the fit. First, it resolved a confusion-axis rotation that the psychophysical loss alone left near zero. For the protan participant, the psychophysical-only fit selected $\hat{\beta}_c \approx +4°$, and its held-out loss equalled that of applying no filter ($\Delta L = +0.01$, improving on it in 3 of 7 folds). The combined fit recovered $+24°$. Second, it reinforced a solution the psychophysical loss had already located. For the deutan participant the psychophysical-only and combined fits shared the same argmin $(6°, -42°)$. The neural term reduced the boundary-saturation rate from 23% to 9.3%. The neural-only fit returned the same sign of $\hat{\beta}_c$ $(-26°)$, corroborating the psychophysical estimate.

Third, it narrowed the HC-resample interquartile range of the argmin in both participants, from $(18°, 6°)$ to $(8°, 2°)$ in the deutan participant and from $(6°, 4°)$ to $(0°, 0°)$ in the protan participant.

The two fitted distortions diverge, with $\hat{\beta}_c = -42°$ in the deutan participant against $+24°$ in the protan participant. The sign of $\hat{\beta}_c$ remained stable when the HC reference set was resampled. Leaving out each of the seven HC reference participants in turn confined $\hat{\beta}_c$ to $[-46°, -38°]$ in the deutan participant and returned an identical argmin on every fold in the protan participant. The fitted signs record displacement at two different anchor directions in a shared hue frame, and the two confusion axes lie $134°$ apart ($\theta_{\text{conf}} = 150°$ and $16°$).

The per-axis magnitudes are not separately identifiable, with 0 of 6 recovery checks surviving FDR correction. What the fits recover is the sign of the dominant confusion-axis term. In the protan participant that sign is basis-dependent. The fit used an RDM loss computed in a PCA-reduced space, and computing the same loss in the SRM-aligned space moves the argmin (Supplementary §4.5). With one participant per subtype, individual and subtype differences remain confounded.

The retinal-plus-gain alternative carries a single degree of freedom, and its cortical gain rescales displacement along the confusion axis alone (Appendix A). It therefore omits displacement away from that axis, including the S-cone intermediate hues where the interpolation deficit concentrated (Section 3.2). Fitting the class registered this misspecification. In both participants the gain settled above 2, which implies cortical overshoot past the undistorted hue and is inconsistent with their confirmed residual deficit (Section 3.5). The single-axis restriction precludes an exact per-person correction, and the correction must instead be derived from the cortical measurement.

4.3 Filter evaluation

The individualized filter left every hue-discrimination pair within the HC range in both participants, whereas the deployed filter rendered two pairs deviant in the protan participant. In the deutan participant both filters removed the baseline threshold deficit. In the protan participant the individualized filter normalized the trending confusion-axis pair green–blue, one of the two pairs the deployed filter left deviant. Color identification accuracy followed the same ordering. Both filters raised it in the deutan participant, whereas in the protan participant it was preserved under the individualized filter and reduced under the deployed filter. This psychophysical advantage appeared in the protan participant only.

The filter's effect on the neural readouts varied by participant and by measure. Adjacent accuracy on hue interpolation separated the two participants, rising in the deutan participant $(0.23 \rightarrow 0.31)$ and falling in the protan participant ($0.14 \to 0.06$). In each participant the early-visual geometry shifted in the direction opposite to interpolation. In the deutan participant both filters moved the geometry away from the HC reference. In the protan participant both filters moved the geometry toward the HC reference. So, that recovery was not specific to the individualized filter. SRM disparity was lower under the individualized filter in the deutan participant and nearly equal between filters in the protan participant. RDM similarity to HC was higher under the deployed filter in both participants. The geometry of both participants remained displaced from the HC reference.

Early-visual geometry and hV4 interpolation lie at different levels of the visual hierarchy and can vary independently. These neural readouts remain inconclusive in a two-case sample. Replication in more individuals within each subtype, on a refined neural endpoint, is required to establish whether either pattern generalizes.

4.4 Limitations

The CVD sample is $N = 2$, one participant per subtype. Subtype was assigned from Ishihara classification plates rather than anomaloscopy, so the confusion-axis angle assumed for each participant carries the uncertainty of that assignment. Single-case statistics (Crawford & Howell, 1998) permit individual-level inference, but population-level claims about deutan or protan subtypes require replication with more participants. Separating a per-person correction from a subtype-average one requires testing several individuals *within* a single subtype. It also requires grading severity on a continuous scale. The plates classify subtype but do not quantify severity. An anomaloscope, which measures the red–green mixture an observer accepts as a match to a reference yellow, would supply both that grading and an independent check on the subtype assignment. The HC reference cohort is likewise small at seven controls. The HC $\|\hat{\beta}\|$ leave-one-out distribution is therefore reported descriptively, as a reference for the magnitude of the CVD estimates rather than as a significance test.

Two of the reported estimates depend on analysis choices. The deutan V2 disparity elevation is significant in the common HC space and falls to a non-significant trend under the symmetric leave-one-subject-out control, whereas the protan V1 elevation is significant under both (Section 3.3). We also report $\hat{\beta}_s$ and $\hat{\beta}_c$ as single values without confidence intervals, because the six runs per participant are too few for stable bootstrap resampling.

The stimulus set was confined to a single locus of constant lightness and chroma ($L^* = 75$, chroma $= 40$). The correction is therefore untested for stimuli that vary in lightness or chroma, including natural scenes. Extending it to those stimuli requires refitting the model at additional levels of both dimensions.

Matching lightness in CIE $L^*a^*b^*$ also equates the stimuli for the standard observer rather than for an individual observer. We measured neither the luminance of the displayed stimuli nor each participant's luminous efficiency. Including both as measured covariates would separate chromatic from luminance contributions to the cortical geometry.

The scanner task diverted attention from color. Participants monitored a letter stream at fixation while the colored disc was present, and no judgment about the disc was required. Attending to color raises response amplitude and tuning gain throughout visual cortex, and in ventral areas it additionally draws colors that share a name closer together (Brouwer & Heeger, 2013). The distortion reported here was measured with attention directed away from color, and a scanner task that requires a color judgment should resolve it more sharply.

The two neural terms in the fitting objective rest on different quantities. L_{LOCO} scores voxel-pattern prediction at hV4 using weights fitted within the participant. L_{RDM} scores the direction of a 28-element vector of pairwise distances at V1 or V2, referenced to the control mean. The interpolation term separated both CVD participants from the control distribution by a large margin. However, it did not enter either selected combination under the three-gate procedure. The same disconnect appears at evaluation, where adjacent accuracy, representational disparity, and RDM similarity to controls moved in different directions under the same filter (Section 3.10). A comprehensive neural index would encompass both interpolation and representational geometry. Sharing that index between the fit and the evaluation is a prerequisite for the joint neural endpoint that a larger study would need.

4.5 Conclusion

We derived a stimulus-space color correction for each of two CVD participants by inverting a model fitted to that participant's own cortical color representation. All displayed hues remained decodable in both participants, whereas the continuous hue geometry departed from the HC reference, at a different cortical area in each. In these two cases the deficit took the form of a hue rotation about two axes of color space, which a uniform attenuation of the chromatic signal does not account for. The fitted parameters differed between the two participants, and inverting each model assigned every displayed hue a single replacement color. Each correction is therefore a per-person transform, not a population-average retinal one.

Larger and systematic studies can now quantify the cortical color distortions of CVD and yield a class of individualized, cortically grounded filters. In the present two-person evaluation the correction's effect varied by participant and by measure. Whether it provides a generalizable improvement in color perception is what those studies would establish.

Data and Code Availability

All analysis and preprocessing code needed to reproduce the reported results is openly available at https://github.com/Transconnectome/colorBlind_analysis [FILL AT SUBMISSION] and archived at Zenodo DOI xx.xxxx/zenodo.xxxxxxx. IN asks for a persistent identifier; make the repository public and mint a release DOI before submission so that reviewers can access it.

The neuroimaging and behavioural data cannot be shared openly. The protocol approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023), and the written consent obtained under it, place all study data in an encrypted store to which the principal investigator controls access, and neither document provides for secondary use by third parties or for deposit in a public repository. Under Article 16(3) of the Seoul National University Research Ethics Guidelines, the de-identified study data are retained for five years and, thereafter, for as long as the principal investigator can maintain custody of them, so that the integrity of the reported analyses can be verified. Requests for access should be directed to the principal investigator (Jiook Cha, connectome@snu.ac.kr) and are granted subject to (i) a formal data sharing agreement between the requesting institution and Seoul National University, (ii) approval of the intended use by the requesting researcher's local ethics committee, and (iii) a short project outline describing that use; a request that falls outside the scope of the original consent additionally requires a new submission to the Seoul National University Institutional Review Board. Co-authorship is not a condition of access. Requests are answered within eight weeks.

Author Contributions

J.K.: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Project administration; Funding acquisition; Writing—original draft; and Writing—review & editing. A.M.C.: Conceptualization; Methodology; Formal analysis; Investigation; Data curation; Project administration; Funding acquisition; and Writing—review & editing. J.S.: Conceptualization (supporting); Investigation; Funding acquisition; and Writing—review & editing. J.C.: Conceptualization; Methodology; Validation; Resources; Supervision; Funding acquisition; and Writing—review & editing.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. No patent has been filed on the correction filter described here, and no author has a financial relationship with a supplier of colour-correction eyewear or display software.

Ethics

The study was approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023) and was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants for being included in the study.

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Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section 3.6). Here we compare it against the retinal-family alternative on the same held-out composite criterion (§2.12), and give the structural reasons the retinal-family class is inadequate for the stimulus-space filter task.

A.1 Retinal-plus-cortical (R+C) fit on the composite criterion

The retinal-plus-cortical model applies the subtype Machado cone-shift $\delta\theta_{\mathrm{Mach}}$ Machado et al., 2009 scaled by a single cortical-gain parameter g, where $\delta\theta_{\mathrm{RC}} = (2-g)\,\delta\theta_{\mathrm{Mach}}$, so that $g=1$ is retinal-only and $g=2$ makes the residual distortion vanish by construction. The form follows the finding that post-receptoral gain amplifies the attenuated retinal signal until the group difference is abolished by V2v and V3v Tregillus et al., 2021, whose gains scale response amplitude rather than a hue displacement. Scored on the same held-out composite as the 2-component model, R+C failed for both participants: for the deutan participant every resample saturated the gain boundary ($g \to 3.0$, 100%), so the fit was rejected as misspecified; for the protan participant the best fit ($g = 2.95$) generalized worse than the adopted 2-component model ($\overline{L}_{\mathrm{test}} = -0.86$ vs. -1.54). Full fits are in Supplementary Table S11 and Section 3.5. A gain $g > 2$ implies cortical overshoot past the undistorted hue, inconsistent with the participants' confirmed residual CVD, so the saturated estimate is a model failure rather than a parameter value.

A.2 Inverse-mapping (pre-image) behavior

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The filter reported in Section 3.9 is that pre-image, evaluated at the eight stimulus hues.

A.3 Why the cortical gain cannot recover the distortion

The cortical gain g rescales the retinal shift uniformly, so for every value of g the R+C family displaces hues along the fixed confusion axis alone. It cannot generate the 2-component model's S-cone-axis term $\beta_s \cos(\theta - 90^\circ)$, a direction that lies 60° from the deutan confusion axis and 74° from the protan one. The observed interpolation deficit concentrates at the S-cone intermediate hues (Section 3.2), a displacement pattern this family cannot represent. The restriction is a property of the model class and holds across fitting criteria.

Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated by rigid-body registration of every volume to its run's middle volume (FSL MCFLIRT) and summarized as framewise displacement (Power et al., 2012), with rotations converted to displacement on a 50 mm sphere. Across the nine analyzed participants mean framewise displacement was 0.32 ± 0.04 mm (HC 0.31 ± 0.04, CVD 0.34 ± 0.05), and 16.2% of volumes exceeded 0.5 mm. These parameters served as a quality-control record only, and the analyzed images were resampled once from native BOLD space into MNI space. A gradient-echo field map was acquired in each session and linked to the functional runs through the BIDS `IntendedFor` field. The analyzed pipeline did not consume it, so the functional data retain susceptibility-induced distortion along the right–left phase-encoding direction. Slow drift was modelled with per-run linear regressors in the general linear model, which constituted the entire nuisance model. Temporal filtering was not applied.

S2. Uncorrected acquisition artifacts

The functional series were acquired with a limited field of view covering occipital cortex. Registration therefore used header-initialized mutual information, which tolerates limited coverage and requires no white-matter boundary. Slice timing, rigid-body motion, and susceptibility distortion were all left uncorrected, and each volume was resampled once by the composed native-to-MNI transform.

Motion sensitivity analysis. Every neural endpoint was recomputed with the six motion parameters and their temporal derivatives added to the second-level design matrix. A control analysis added the same twelve regressors after circularly shifting each within run. The shift preserves the autocorrelation and spectrum of the regressors while destroying their temporal alignment with the data, and therefore separates the variance inflation caused by adding twelve regressors from the removal of motion-aligned variance.

Table S1: Procrustes disparity under the symmetric leave-one-subject-out reference, across the three preprocessing arms. Crawford–Howell one-tailed p.

Participant	ROI	Original	Motion regression	Shifted control
Protan	V1	0.045	0.022	0.031
Deutan	V2	0.116	0.300	0.044

The protan V1 elevation survives motion regression and strengthens under it. The deutan V2 value, already non-significant under the leave-one-subject-out reference, falls further. The shifted control retains the deutan elevation, so the attenuation under motion regression does not follow from the loss of degrees of freedom or from collinearity among the added regressors.

Split-half RDM reliability, averaged over the four regions, gives the same ordering. Taking the difference between the motion-regression arm and the shifted control as the net effect of removing motion-aligned variance, the deutan participant loses 0.338, which is 2.5 times the next largest value in the cohort (0.134), whereas the protan participant is unaffected ($+0.002$).

Head motion was not locked to the stimulus sequence in any participant. Regressing the motion parameters on the eight-color design matrix and comparing the resulting R^2 to a null built from circularly shifted onsets gave z between -0.07 and $+0.08$ and p between 0.34 and 0.63 across the ten participants.

Susceptibility distortion. Distortion displaces signal along the phase-encoding direction, which ran right to left. Within a participant it is constant across conditions, so it cannot generate condition-specific differences among the eight colors. Its cost is spatial, and it reduces the precision with which atlas-defined regions are assigned to functional voxels.

Coregistration stability. Registration of the occipital BOLD volume to the whole-head anatomical image was probed in two ways. Across the runs of a single session, where the fitted transform should be identical, the mean pairwise displacement of the solution ranged from 0.9 to 4.2 mm depending on participant and session. Removing facial voxels from the anatomical image, which leaves the brain unchanged, moved the solution by 1.9 mm in the deutan participant and 9.4 mm in the protan participant. The mutual-information optimum is therefore shallow when the field of view is restricted to occipital cortex. The transform is fixed within a run, so it cannot produce differences among the eight colors that a run contributes. Registration variability instead enters as run-to-run noise, which averaging over the six runs attenuates.

Limits. Motion regression removes variance temporally aligned with the estimated parameters. It does not reconstruct the data that volume realignment would have produced, so this analysis bounds the contribution of motion-aligned variance rather than establishing what a fully corrected pipeline would yield. Realignment and distortion correction of limited-FOV series are both nontrivial, since rigid-body estimation is less constrained when little of the head is imaged. Acquisitions of this type would benefit from correction validated for partial coverage.

S3. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S4. Quality Control

ROI coverage averaged 84.3% (SD 21.7%) across the ten scanned participants, and 99.6% of ROI voxels carried reliable stimulus-evoked responses. Coverage was computed as the intersection of the Wang atlas ROIs (V1, V2, V3, hV4) with each participant's BOLD brain mask in MNI space. Coverage reached 30.8% in sub-07, the lowest value in the sample, and all participants entered the downstream analyses.

S5. Mean Activation Analysis

Activation-level metrics were compared between the healthy controls and each CVD participant on the amplitudes before SRM alignment. The metrics were mean activation magnitude (mean $|\beta|$), signal-to-noise ratio, color modulation depth, run-to-run reliability, and spatial variance, each evaluated at V1, V2, V3 and hV4. Every comparison is a single-case test against the seven controls (Crawford and Howell (1998), two-tailed), the procedure used for the geometric and interpolation endpoints.

Both CVD participants fell inside the control distribution on every metric at every ROI. For mean $|\beta|$ the largest deviation was $d_{cc} = +1.61$ in the deutan participant at V3 ($p = 0.182$), and every value satisfied $p \geq 0.182$. For color modulation depth the largest deviation was $d_{cc} = +2.16$ in the deutan participant at V2 ($p = 0.090$). The largest deviations on both metrics were positive rather than negative, which is the opposite of the direction a uniform attenuation predicts. Voxel-level color selectivity was substantial in both groups (one-way ANOVA across voxels, $F > 4$, $p < 0.001$ in V1–V3). With seven controls a two-tailed test at $\alpha = 0.05$ requires a deviation of 2.62 control standard deviations, and its power against a true deviation of 2 SD is approximately 0.35. These tests therefore rule out a substantial uniform attenuation rather than establishing equality.

Activation metrics and SRM disparity varied independently across participants, with Pearson r from -0.29 to $+0.51$ over 20 tests and all $p \geq 0.130$. The HC–CVD difference therefore resides in the representational pattern rather than in response amplitude. The cortical color response retains its strength in CVD while its geometry departs from the control reference.

S6. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined by leave-one-subject-out cross-validation over the candidate values 2 to 6 (Cunningham & Yu, 2014). On each of the seven folds, one HC participant was withheld from SRM training and three quality metrics were computed for every candidate. RDM split-half reliability was the Spearman correlation between representational dissimilarity matrices built from the odd runs 1, 3, and 5 and the even runs 2, 4, and 6 of the withheld participant in SRM space. The remaining two metrics were the cross-subject RDM correlation and the reconstruction error. Candidates were ranked within each fold and metric, and the candidate with the best mean rank was selected. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S7. Generalized Cross-Validation for Ridge Regression

This penalty applies to the encoding direction alone. Hue decoding uses the unregularized pseudoinverse, because a ridge penalty on the channel-to-voxel weights degrades the decoding readout (§S8). The regularization parameter α for ridge regression was selected by generalized cross-validation (Golub et al., 1979), which approximates leave-one-out cross-validation at a fraction of its cost. We implemented it directly from the singular value decomposition of the channel design matrix, searching the grid $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$. A single α was selected per fit by minimizing the GCV score:

$$\text{GCV}(\alpha) = \frac{\overline{\|Y - \hat{Y}\|_2^2}}{(1 - \text{tr}(H)/n)^2} \tag{7}$$

where C denotes the channel design matrix, Y the observed responses of all voxels in the region, $\hat{Y}$ the ridge prediction, $H = C(C^T C + \alpha I)^{-1} C^T$ the hat matrix, and n the number of training samples. The squared residuals are averaged over both samples and voxels, so the selected penalty is shared across the voxels of a region.

S8. Cross-Validation Procedures

Leave-one-run-out (LORO). LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B, C) were held out as the test set, W was trained on the remaining five runs. The held-out run's color patterns were then decoded by nearest-neighbor matching in channel-output space using the OLS pseudoinverse weights (Methods §2.7).

Leave-one-color-out (LOCO). LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors. The held-out color was then decoded from voxel responses using the OLS pseudoinverse weights, yielding the adjacent-accuracy vulnerability profile. The complementary voxel-response prediction (ridge-GCV encoding) enters only the fitting loss L_{LOCO}, not the decoding metric (Methods §2.7).

Cross-run alignment and leakage control. The run-level Procrustes alignment (Methods, ROI definition and response estimation) maps runs 2–6 onto a fixed run-1 reference by an orthogonal rotation that uses no stimulus labels. Because the alignment target is a single fixed run rather than a jointly fitted shared space, the held-out run or color does not define the alignment frame. Re-estimating the alignment inside each fold raised forward-encoding LORO accuracy from 0.545 to 0.578. Leakage in the fixed-reference pipeline would predict the opposite direction, so the reported estimate is not inflated by the alignment step. That variant also replaces the run-1 reference with the training-run mean, so it bounds the leakage question rather than isolating the nesting itself.

Leave-one-subject-out (LOSO). LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants' shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S9. Evaluation Metrics

Color decoding metrics. Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90°, expected from a uniform random predictor on a circle), and (2) eight-way classification accuracy, obtained by assigning the decoded hue to the nearest stimulus hue, equivalently a $\pm 22.5°$ bin around each of the eight colors (chance = 12.5% = $1/8$).

Voxel prediction metrics. Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical significance. Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols and Holmes, 2002).

S10. Comparison with alternative decoders

The forward encoding model was adopted for its interpolation behaviour rather than for its classification accuracy. We compared six decoders on the same SRM-aligned amplitudes under both cross-validation schemes. Classification separates the decoders in one direction and interpolation separates them in the other, and the interpolation result is the one the present analyses depend on.

Linear discriminant analysis classified held-out runs most accurately at every ROI (table S2). Its healthy-control mean reached 0.878 at V1 against a chance level of 0.125, with the support vector machine next at 0.777 and the forward encoding model at 0.542. The ordering held at V2, V3 and hV4. Four of the six decoders exceeded chance at every ROI, so eight-way classification is recoverable by several readouts and does not by itself favour any one of them.

Interpolation to a held-out color reverses the ordering at hV4, the ROI on which the main analyses rest. The forward encoding model reached an adjacent accuracy of 0.470 there, against a chance level of 0.25 for its continuous-hue output (table S3). It is the only decoder whose healthy-control mean exceeds its own chance level, and it does so at V1 (0.360), V2 (0.283) and hV4, with the widest margin at hV4. The classifiers stay below their 0.375 chance level at every ROI, and ridge and kernel ridge stay below 0.25 at every ROI. The analytic chance level compares a decoder against a uniform prediction. The main-text criterion is stricter, since the color-label permutation null preserves the covariance among voxels, and hV4 is the region that meets it (§3.2). This pattern follows from the model form: the forward encoding model represents a hue as activation over six tuned channels and can therefore evaluate a hue it never saw during training, whereas a classifier trained on seven discrete labels has no output unit for the eighth.

The correlation readout entered this comparison as the incumbent rather than as its outcome. It was implemented for the Phase-1 forward model in January 2026 and specified in the analysis plan in February 2026, both before the runs reported here. The comparison therefore verifies that the readout was not the limiting factor, and the wording in the main text reflects that scope.

Two rows require care. Kernel ridge regression returned an adjacent accuracy of exactly 0.000 in every participant, ROI and alignment, and the multilayer perceptron returned exactly the chance value with zero variance across participants, which is the signature of a constant prediction. Neither is treated as a competitive score. Three further variants are outside this table: the ensemble decoders were retired from the pipeline, the two hybrid decoders were run for interpolation only, and four alternative readouts of the forward encoding model vary the readout rather than the decoder.

S11. Alignment Robustness for the Within-Subject Readouts

LORO and LOCO are trained and tested within a single participant, so the main text computes them on the Procrustes-aligned amplitudes, which retain that participant's voxels at full resolution. SRM is applied only where a comparison spans participants. table S4 repeats both readouts in the SRM-aligned space as a robustness check.

The two spaces agree on the pattern that carries the Results. Classification stays far above the 0.125 chance level for both CVD participants at every region in both spaces, with the lowest cell at 0.312. Interpolation at hV4 remains reduced in both participants in both spaces, and the ordering of the two participants is preserved. Procrustes alignment yields the higher control-mean accuracy at all four regions for classification and at three of the four for interpolation, and it is the higher of the two spaces in 26 of the 36 participant-by-region classification cells. This is consistent with the dimensionality reduction in SRM discarding response variance that a within-subject readout can otherwise use.

The spaces differ at V1 classification, where SRM places both CVD participants above the control mean ($d_{cc} = +0.59$ and $+0.79$) while Procrustes places them slightly below ($d_{cc} = -0.25$ for both). No single-case classification comparison reaches significance in either space.

Table S2: Eight-way classification accuracy under leave-one-run-out cross-validation, SRM-aligned amplitudes, chance $= 0.125$. Healthy-control mean $\pm$ SD ($n = 7$), with the two CVD participants listed separately. Bold: highest control mean in the ROI.

ROI	Decoder	HC (mean $\pm$ SD)	Deutan	Protan
V1	LDA	$\mathbf{0.878 \pm 0.050}$	1.000	0.854
	SVM	0.777 ± 0.062	0.979	0.771
	Forward encoding	0.542 ± 0.106	0.604	0.625
	Ridge	0.399 ± 0.054	0.354	0.250
	Kernel ridge	0.393 ± 0.064	0.271	0.292
	MLP	0.128 ± 0.008	0.125	0.125
V2	LDA	$\mathbf{0.830 \pm 0.047}$	0.917	0.854
	SVM	0.759 ± 0.136	0.875	0.833
	Forward encoding	0.545 ± 0.077	0.458	0.438
	Ridge	0.372 ± 0.097	0.458	0.250
	Kernel ridge	0.330 ± 0.066	0.313	0.250
	MLP	0.149 ± 0.031	0.125	0.125
V3	LDA	$\mathbf{0.726 \pm 0.064}$	0.750	0.771
	SVM	0.631 ± 0.144	0.813	0.667
	Forward encoding	0.449 ± 0.065	0.375	0.313
	Ridge	0.173 ± 0.068	0.292	0.229
	Kernel ridge	0.158 ± 0.052	0.208	0.333
	MLP	0.122 ± 0.008	0.125	0.125
hV4	LDA	$\mathbf{0.658 \pm 0.092}$	0.813	0.729
	SVM	0.598 ± 0.085	0.813	0.667
	Forward encoding	0.423 ± 0.072	0.354	0.354
	Ridge	0.319 ± 0.114	0.146	0.250
	Kernel ridge	0.277 ± 0.093	0.042	0.188
	MLP	0.125 ± 0.000	0.125	0.125

S12. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-HC-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i}, yielding disparity scores $d(\text{HC}_i, R_{-i})$ and $d(\text{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

Table S3: Adjacent accuracy under leave-one-color-out cross-validation, SRM-aligned amplitudes. Healthy-control mean ± SD ($n = 7$), with the two CVD participants listed separately. Chance depends on the output space of the decoder. Forward encoding, ridge and kernel ridge return a continuous hue, so a uniform prediction lands within the $45°$ tolerance on 91 of 360 draws and chance is 0.25. The LDA, SVM and MLP classifiers return one of the eight stimulus hues, for which the tolerance admits 3 of 8 and chance is 0.375. The constant-output MLP sits at exactly 0.375 at V1 and V2, which confirms the classifier value. Bold: the largest margin above the applicable chance level. Kernel ridge and the MLP are constant-output rows, noted in the text.

ROI	Decoder	HC (mean ± SD)	Deutan	Protan
V1	Forward encoding	0.360 ± 0.098	0.313	0.250
	LDA	0.268 ± 0.074	0.375	0.167
	SVM	0.226 ± 0.078	0.375	0.229
	Ridge	0.039 ± 0.053	0.146	0.000
	MLP	0.375 ± 0.012	0.375	0.375
	Kernel ridge	0.000 ± 0.000	0.000	0.000
V2	Forward encoding	0.283 ± 0.101	0.333	0.063
	LDA	0.327 ± 0.088	0.313	0.417
	SVM	0.318 ± 0.108	0.250	0.396
	Ridge	0.054 ± 0.083	0.021	0.000
	MLP	0.375 ± 0.000	0.396	0.375
	Kernel ridge	0.000 ± 0.000	0.000	0.000
V3	Forward encoding	0.220 ± 0.100	0.354	0.125
	LDA	0.220 ± 0.081	0.271	0.333
	SVM	0.208 ± 0.103	0.250	0.229
	Ridge	0.000 ± 0.000	0.000	0.000
	MLP	0.250 ± 0.000	0.250	0.250
	Kernel ridge	0.000 ± 0.000	0.000	0.000
hV4	Forward encoding	$\mathbf{0.470 \pm 0.130}$	0.271	0.104
	LDA	0.280 ± 0.144	0.146	0.229
	SVM	0.253 ± 0.126	0.167	0.208
	Ridge	0.000 ± 0.000	0.000	0.000
	MLP	0.250 ± 0.000	0.250	0.250
	Kernel ridge	0.000 ± 0.000	0.000	0.000

Common-space vs. symmetric-LOSO estimates. The geometric-disparity tests (§3.3) are reported primarily in the common HC-trained SRM space, in which the seven HC train the shared space and each subject of interest is projected in; because the held-out HC reference is in-sample whereas the projected CVD participant is out-of-sample, we additionally computed a symmetric leave-one-subject-out (LOSO) estimate in which every HC and CVD subject alike is projected by SVD into a space trained on the remaining six HC (Methods, §2.9). Both use the Crawford and Howell, 1998 one-tailed single-case

Table S4: Forward-encoding readouts in the two alignment spaces. LORO is eight-way classification accuracy (chance 0.125); LOCO is adjacent accuracy (analytic chance 0.25). HC is the mean over $n = 7$. The main text reports the Procrustes columns.

	LORO classification			LOCO interpolation		
Region	HC	deutan	protan	HC	deutan	protan
Procrustes-aligned (main text)						
V1	0.580	0.562	0.562	0.393	0.437	0.188
V2	0.607	0.521	0.562	0.357	0.271	0.104
V3	0.574	0.396	0.458	0.339	0.375	0.208
hV4	0.488	0.375	0.375	0.456	0.250	0.125
SRM-aligned (robustness)						
V1	0.542	0.604	0.625	0.360	0.312	0.250
V2	0.545	0.458	0.438	0.283	0.333	0.062
V3	0.449	0.375	0.312	0.220	0.354	0.125
hV4	0.423	0.354	0.354	0.470	0.271	0.104

t ($df = 6$). Table S5 reports both estimates for all ROIs. The protan participant's V1 elevation is significant under both estimators. The deutan participant's V2 elevation is significant in the common space and does not survive under LOSO ($p = 0.116$, $d_{cc} = 1.42$). V3 in that participant behaves the same way under both estimators ($p = 0.052$ and $p = 0.143$), so the two regions are not separable in the deutan participant.

Table S5: Procrustes disparity, Crawford–Howell one-tailed single-case t ($df = 6$) against the HC reference, in the primary common HC-trained SRM space and under the symmetric LOSO control, with the case-control effect size $d_{cc} = t\sqrt{8/7}$ ($n = 7$) given for each estimator. Bold: $p < 0.05$.

		Common space			Symmetric LOSO		
Subject	ROI	t	p	d_{cc}	t	p	d_{cc}
Deutan	V1	1.10	0.157	1.18	0.48	0.323	0.51
	V2	$\mathbf{2.11}$	$\mathbf{0.040}$	2.26	1.33	0.116	1.42
	V3	1.92	0.052	2.05	1.17	0.143	1.25
	hV4	0.23	0.411	0.25	0.07	0.474	0.07
Protan	**V1**	$\mathbf{3.48}$	$\mathbf{0.007}$	3.72	$\mathbf{2.02}$	$\mathbf{0.045}$	2.16
	V2	0.99	0.181	1.06	0.77	0.234	0.82
	V3	0.09	0.466	0.10	0.06	0.479	0.06
	hV4	1.13	0.150	1.21	0.80	0.228	0.86

S13. Validity of the geometric comparison

Holding the shared projection fixed is required for the color-label permutation to carry power. Re-estimating the projection at each permutation lets the re-fit absorb the shuffle. We tested this on the healthy controls, in whom color structure is established. Under re-estimation the procedure detected 0 of 7 controls at V1, V2 and V3, and 0 of 6 at hV4, with mean permutation z near zero in every ROI. Freezing the projection recovered detection, reaching 5 of 7 at V3 and 2 to 3 of 7 elsewhere (table S6). Every color-specificity result below therefore uses the frozen projection.

Table S6: Color-label permutation on the healthy controls, used as a positive control. Detection counts and mean permutation z under a projection re-estimated at each permutation and under a frozen projection. Lower z indicates a better identity correspondence than the shuffled null.

		Re-estimated		Frozen	
ROI	n	detected	mean z	detected	mean z
V1	7	0	-0.04	2	-1.58
V2	7	0	-0.04	3	-1.00
V3	7	0	-0.78	5	-2.39
hV4	6	0	$+0.06$	2	-1.07

Both CVD participants carried color-specific geometry, at different ROIs. The deutan participant reached $p = .002$ at V2, the lowest V2 value among the nine participants, and $p = .024$ at V3. The protan participant reached $p = .001$ at V3 and $p = .013$ at V2, while V1 returned $p = .758$. Across the 35 participant-by-ROI cells, 16 fell below $.05$ against a chance expectation of 1.8, and 7 survived Benjamini–Hochberg correction over all 35 cells (table S7). Two of the surviving cells belong to the CVD participants, the deutan V2 cell ($q = .018$) and the protan V3 cell ($q = .012$). V3 carried color specificity most broadly, with 5 of its 9 cells surviving correction.

Motion regression broadened the pattern rather than removing it. The count of cells surviving correction rose from 7 to 15, and the deutan V2 cell held at $q = .025$ while the protan V3 cell held at $q = .012$. Two readings of this increase remain open. Motion may contribute label-independent variance that masks color structure, or the twelve added regressors may reshape the residual variance in a way that favours the observed correspondence. The circular-shift control that separates these accounts was applied to the disparity endpoint (S2) and remains to be extended to the permutation reported here.

Disparity and color specificity address different questions. Disparity measures the distance between two geometries after optimal rotation and stays invariant to how the colors map onto one another. The permutation asks whether the identity mapping between a participant's colors and the reference colors outperforms a shuffled mapping. A participant can therefore sit far from the reference in disparity

Table S7: Color-label permutation under the frozen projection, one cell per participant and ROI. Entries give p_{perm} with the Benjamini–Hochberg q in parentheses, corrected across all 35 cells. Bold: $q < 0.05$. HC 7 lacks sufficient hV4 coverage.

Participant	V1	V2	V3	hV4
HC 1	0.088 (0.162)	0.020 (0.064)	0.062 (0.120)	0.393 (0.443)
HC 2	**0.001** (0.012)	0.016 (0.056)	**0.004** (0.023)	0.136 (0.207)
HC 3	0.109 (0.181)	0.520 (0.552)	0.255 (0.329)	0.272 (0.329)
HC 4	0.152 (0.221)	0.014 (0.054)	**0.001** (0.012)	0.041 (0.090)
HC 5	0.057 (0.117)	0.773 (0.773)	0.034 (0.079)	0.031 (0.079)
HC 6	0.407 (0.445)	0.290 (0.338)	**0.009** (0.045)	0.220 (0.296)
HC 7	0.034 (0.079)	0.159 (0.222)	**0.004** (0.023)	—
Deutan	0.105 (0.181)	**0.002** (0.018)	0.024 (0.070)	0.273 (0.329)
Protan	0.758 (0.773)	0.013 (0.054)	**0.001** (0.012)	0.129 (0.205)

while the identity mapping remains the best correspondence available, or sit close to it while a shifted correspondence fits better.

The protan participant's V1 correspondence is displaced by one hue step. Rotating the color labels by $45°$ lowered Procrustes disparity from 1.037 to 0.788, a gain of 24.0%. We referred that gain to the distribution of shift gains across the seven healthy controls ($3.5 \pm 5.9\%$), which are computed under the same minimum-over-eight rule and therefore carry the same selection bias. The protan gain exceeded that distribution (Crawford–Howell $t = 3.22$, $p = .009$, $d_{cc} = 3.44$), and the rotated geometry sat below the control mean disparity of 0.839 ± 0.087. The same rotation raised the second-order RSA similarity from $\rho \approx 0.00$ to $\rho = +0.52$, against a control mean of 0.45, exceeding its own selection-bias null ($z = +5.02$, $p = .002$, $p_{\mathrm{adj}} = .032$). Motion regression preserved both effects (disparity gain 19.2%, $t = 2.43$, $p = .025$). The deutan participant showed the identity mapping as optimal at V1, V2 and V3.

The evidence supports the existence of a color rearrangement that restores control-level similarity in the protan participant at V1. Which rearrangement it is remains open. Under the SRM basis a $225°$ shift ($+0.50$) sits close to the $315°$ optimum ($+0.52$), and under the PCA basis the optimum falls at $135°$ ($p = .048$, which does not survive correction).

S14. Alignment-independent checks on the disparity measure

Disparity tracks the same subject-level quantity when the alignment method changes. We recomputed each participant's distance from the healthy-control mean with two procedures that use no shared response model, then correlated the result with the SRM disparity used in the main analyses. Crossnobis distance in native voxel space agreed at $r = 0.632$ pooled across ROIs ($p < .001$), and PCA distance agreed at $r = 0.780$ ($p < .001$). Agreement was strongest at V1 and V2, the two ROIs carrying the single-case elevations (table S8).

Cross-validated Mahalanobis distance reproduces the ordering without any alignment step. Following Walther et al. (2016), we computed an 8×8 crossnobis distance matrix per participant in native voxel space, cross-validated over the 15 run pairs, with Ledoit–Wolf shrinkage of the noise covariance. Control-to-control RDM similarity exceeded control-to-CVD similarity at V1 by 0.104 (permutation $p = .120$), and the three remaining ROIs gave differences below 0.05. The convergent correlation with SRM disparity reached $r = 0.833$ at V1 ($p = .005$) and $r = 0.733$ at V2 ($p = .025$).

A second alignment algorithm gives the same subject ordering. We replaced SRM with PCA at matched dimensionality, with and without a subsequent canonical-correlation alignment, and computed Procrustes disparity over all participant pairs. Group-level effects were small under both variants, with Hedges' g from -0.13 to $+0.40$ under PCA and from -0.13 to $+0.16$ under PCA-CCA. Pairwise alignment fits each pair independently and therefore carries more noise than a jointly fitted shared space, which accounts for the size of these effects. Subject-level agreement with SRM disparity was substantial under both (PCA pooled $r = 0.780$, $p < .001$; PCA-CCA pooled $r = 0.544$, $p < .001$).

The shared space reconstructs the CVD data at least as well as the control data, which places reconstruction quality outside the set of explanations for the elevated disparity. Under the symmetric leave-one-subject-out framework, variance explained was higher in CVD than in controls at all four ROIs, and the V2 difference was the largest (controls 0.331, CVD 0.416, $g = -1.26$). Variance explained and disparity varied independently across participants (pooled $r = -0.214$, $p = .211$), so the two indices track separate properties. High reconstruction quality together with high disparity describes a strong signal held in a different arrangement.

These checks establish that disparity is a property of the data rather than of the alignment procedure. They concern the measurement, and whether the deviation is specific to color is treated in S12. Group-level contrasts at this sample size carry wide intervals, so the argument rests on the subject-level agreement between three distances computed under three different alignment procedures.

S15. Session-1 Hue-Discrimination Thresholds

table S10 lists the unfiltered Session-1 threshold for every tested hue pair. The threshold is the staircase level t, the fraction of the pair's full CIE LCh separation at which the two discs were discriminated

Table S8: Convergent validity of SRM disparity with two alignment-independent distances, Spearman r over the nine analysed participants. Crossnobis distance uses native voxel space with no alignment step. PCA and PCA-CCA replace the shared response model with a pairwise alignment at matched dimensionality. Bold: $p < 0.05$.

Distance	V1	V2	V3	hV4	Pooled
Crossnobis (voxel space)	**0.833**	**0.733**	0.550	0.300	**0.632**
PCA	0.633	**0.850**	0.333	0.533	**0.780**
PCA-CCA	0.483	0.433	0.100	-0.067	**0.544**

Table S9: Variance explained by the shared response model under the symmetric leave-one-subject-out projection, in which controls and CVD participants are projected identically. Hedges' g is signed control minus CVD, so negative values indicate better reconstruction of the CVD data.

ROI	k	Controls	CVD	g
V1	4	0.352	0.407	-0.40
V2	4	0.331	0.416	-1.26
V3	3	0.250	0.314	-0.66
hV4	3	0.225	0.253	-0.39

(§2.3), so smaller values indicate finer discrimination. The ratio γ divides the participant's threshold by the healthy-control mean, and z expresses the same difference in control standard deviations over the seven controls. Pairs are grouped by the axis each was chosen to probe, and that assignment was fixed before any threshold was collected.

The elevations align with subtype. The deutan participant exceeded $z = +4$ on both deutan-axis pairs and on the S-cone pair, whereas the protan participant exceeded the control range only on the protan-axis pair green–blue. Neither participant was elevated on the $180°$ control pair, which argues against a general loss of sensitivity. These are the values entering the psychophysical loss atoms of §2.10; single-case tests of the same pairs are reported with the filter evaluation, where they serve as the pre-filter baseline (§3.10).

S16. Comparison with Retinal-Family Distortion Models

The retinal-family (R+C) model was evaluated alongside the 2-component model under the same grid-search and held-out composite criterion (§2.12).

Model. The retinal-plus-cortical (R+C) model is defined in §2.10: the subtype Machado cone-shift $\delta\theta_{\mathrm{Mach}}$ Machado et al., 2009, fixed by the confusion axis, scaled by a single cortical-gain parameter g

Table S10: Session-1 hue-discrimination thresholds without a filter. HC is the mean $\pm$ SD over the seven healthy controls. γ is the ratio to the HC mean and z the deviation in HC standard deviations. Bold: $|z| \geq 2$.

		HC ($n = 7$)		Deutan			Protan		
Axis probed	Pair	mean	SD	t	γ	z	t	γ	z
Deutan confusion	orange–yellow	0.278	0.135	0.840	$\mathbf{3.02}$	$\mathbf{+4.15}$	0.193	0.69	-0.63
	yellow–green	0.090	0.044	0.278	$\mathbf{3.10}$	$\mathbf{+4.32}$	0.128	1.42	$+0.87$
Protan confusion	green–blue	0.079	0.032	0.078	0.98	-0.06	0.155	$\mathbf{1.95}$	$\mathbf{+2.36}$
	cyan–magenta	0.042	0.017	0.040	0.95	-0.13	0.038	0.89	-0.28
S-cone	yellow–purple	0.022	0.006	0.063	$\mathbf{2.87}$	$\mathbf{+6.70}$	0.028	1.26	$+0.94$
	blue–purple	0.164	0.093	0.120	0.73	-0.48	0.148	0.90	-0.18
	red–orange	0.124	0.072	0.063	0.50	-0.85	0.075	0.61	-0.68
Control	red–cyan	0.034	0.015	0.015	0.45	-1.23	0.015	0.45	-1.23
Mean $\lvert z\rvert$ over the eight pairs	Deutan 2.24	Protan 0.90							

($\delta\theta_{\mathrm{RC}} = (2-g)\,\delta\theta_{\mathrm{Mach}}$; $g = 1$ is retinal-only, $g = 2$ exact cortical cancellation; Tregillus et al., 2021). It was fit by the same grid-search and held-out composite criterion as the 2-component model (§2.12), so the two model classes are directly comparable on $\overline{L}_{\mathrm{test}}$.

Fits (Table S11). The retinal-plus-cortical model reached the gain boundary in both participants. Anchor choice matters for the protan participant, where boundary saturation ranged from 0% to 100% across the three $\Delta\lambda$ values, so the reported fit is the anchor that generalized best. For the deutan participant, 100% of resample solutions saturated at $g \to 3.0$, and the fit was rejected at the boundary-saturation gate as model misspecification (§3.5). For the protan participant, 41% of solutions saturated at $g = 2.95$, which passed that gate, and the held-out composite favoured the 2-component fit ($\overline{L}_{\mathrm{test}} = -1.54$ against -0.86). A gain above 2 implies cortical overshoot past the undistorted hue, which the participants' confirmed residual CVD excludes. The single-axis degree-of-freedom deficit accounts for this behaviour (§3.5).

Relation to ΔRDM and LOCO. Fitting a retinal-family model directly against ΔRDM reproduced the same boundary behaviour in both participants. For the adopted 2-component model, the ΔRDM cosine L_{RDM} enters the selected composite loss (§2.12). The forward-encoding LOCO family remains reserved for diagnostic decoding and entered neither adopted combination.

Table S11: Distortion-model fits for both CVD participants, all scored on the same held-out composite test-loss ($\overline{L}_{\text{test}}$, lower is better; §2.12; the neural RDM atom in each adopted fit is V2 for the deutan and V1 for the protan participant). The 2-component model is shown at its adopted fit. The retinal-plus-cortical (R+C) fits are reported for comparison: the deutan R+C was rejected at the boundary-saturation gate (§3.5), and the protan R+C generalized worse than the 2-component fit.

Participant	Model	Fitted parameters	Held-out fit
Deutan	2-component	$\beta_s = 6°,\ \beta_c = -42°$	$\overline{L}_{\text{test}} = -2.36$
Protan	2-component	$\beta_s = 2°,\ \beta_c = +24°$	$\overline{L}_{\text{test}} = -1.54$
Deutan	R+C	$g \to 3.0$ (100% saturated)	rejected (Gate 2)
Protan	R+C	$g = 2.95$ (41% saturated)	$\overline{L}_{\text{test}} = -0.86$

S17. HC leave-one-out magnitude-anchor distribution

Both CVD estimates fall inside the corresponding HC range, which places this section's role as a descriptive magnitude anchor rather than a test of CVD-versus-HC specificity. The deutan estimate is $\|\hat{\beta}\| = 42.4°$ against an HC range of $30.5°$ to $58.1°$ (mean $49.1°$). The protan estimate is $24.1°$ against $23.4°$ to $55.5°$ (mean $35.7°$), close to the lower bound of that range. Parameter magnitude alone therefore places both CVD cases within the HC distribution.

The production loss differs between participants, the deutan combination being $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ and the protan combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$, so each anchor was computed under that participant's own loss. Each of the $n = 7$ HC participants was refit as a pseudo-CVD case with the remaining HC pool as reference.

S18. Identifiability checks

Effective parameter uncertainty is approximately $20°$ on β_s and $25°$ on β_c under the PCA-basis loss, read from the median and interquartile range of the origin-recovery null (Test 2a). The production argmin therefore functions as a descriptive embedding of the CVD-specific distortion direction rather than as a point estimate with physiological magnitude. Four pre-specified checks establish this bound and delimit the claim. Test 1 assesses voxel-level parameter recovery, Test 2a measures the algorithm's noise floor, and Tests 2b and 2c assess specificity to CVD data. Each test was run on the two production candidates (S08-robust, S09-primary), using PCA-basis voxel synthesis (spatial PCA(20) on SRM-projected amplitudes) as the feature space. Benjamini–Hochberg correction across the six test-bearing checks (2 candidates $\times$ 3 tests, $\alpha = 0.05$) returned no result below threshold.

Test 1 — Voxel-level parameter recovery. Synthetic fMRI data were generated at the production argmin (PCA-basis; 7 HC donors $\times$ 20 noise draws $= 140$ samples per candidate) with GT-consistent JND: psychophysical thresholds were scaled to match the distortion magnitude at the ground-truth (β_s, β_c). The same fitting pipeline was re-run on each synthetic sample. Pass criterion: fraction within $10°$ on both axes ($f_{10°}$) ≥ 0.5 and $|$bias$| < 10°$ on both axes.

Candidate	bias $(\beta_s,\ \beta_c)$	$f_{10°}$	Verdict
S08-robust $(6°, -42°)$	$(+16°,\ -4.7°)$	0.26	FAIL
S09-primary $(2°, +24°)$	$(+11°,\ -27°)$	0.14	FAIL

Table S12: Test 1 results. Recovery reached $f_{10°} = 0.26$ and 0.14 against a pre-specified criterion of 0.5. The non-dominant axes ($|\hat{\beta}_s| \leq 6°$ for both participants) fall below the effective parameter uncertainty established by Test 2a and are not recoverable. For S08-robust, the dominant axis ($|\hat{\beta}_c| = 42°$, bias $= 4.7°$) shows lower bias than the non-dominant axis ($16°$), consistent with partial magnitude recoverability where the axis magnitude exceeds the noise floor.

Test 2a — Origin-recovery null. Synthetic fMRI data were generated at GT $= (0, 0)$ using real HC JND (no synthetic psychophysical signal; PCA-basis). At the null, real HC JND carries no CVD-consistent signal, making this a noise floor measurement free of synthesis design contamination. Pass criterion: $|$bias$| < 5°$ on both axes.

Candidate	$\lvert\hat{\beta}_s\rvert_{\text{med}}$ (IQR)	$\lvert\hat{\beta}_c\rvert_{\text{med}}$ (IQR)	β_c p95	$f_{10°}^{\text{origin}}$
S08-robust	$22°\ (40)$	$26°\ (10.5)$	$44°$	0.00
S09-primary	$16°\ (17.5)$	$24°\ (9)$	$48°$	0.00

Table S13: Test 2a results. $f_{10°}^{\text{origin}} = 0/140$ for both candidates: no sample was recovered within $10°$ of the origin under a null ground truth. Effective parameter uncertainty (PCA-basis): $\sim 20°$ (β_s) and $\sim 25°$ (β_c).

Test 2b — HC pseudo-CVD specificity. Each HC participant's real amplitude data were substituted as a fake CVD participant (7 carriers per candidate, deterministic). The production CVD fit distance from the origin was ranked among the 7 HC pseudo-CVD distances (rank$_{\text{dist}}$, one-sided high). Pass criterion: rank $= 1.0$ (CVD fit exceeds all HC distances). This test is descriptive only and was not used as a selection criterion.

Candidate	rank$_{\text{dist}}$	Verdict
S08-robust	0.875	FAIL
S09-primary	0.875	FAIL

Table S14: Test 2b results. Neither candidate exceeds all HC pseudo-CVD distances.

Test 2c — Color-label permutation. CVD trial labels were randomly shuffled ($N = 1000$ permutations); HC data were held fixed. Pass criterion: $p_{\text{perm}} < 0.05$ (one-sided, lower = better).

Candidate	Real loss	Perm 5% cut	p_{perm}
S08-robust	-2.892	-3.136	0.167
S09-primary	-1.681	-3.053	0.471

Table S15: Test 2c results. Neither candidate reaches significance against the label-permutation null.

FDR summary and basis caveat. After FDR correction across 6 tests (2 candidates × 3 main tests; BH $\alpha = 0.05$), no test reaches significance. The production argmin is treated as a descriptive embedding of the CVD-specific distortion direction throughout.

All results above use the PCA-basis loss. For the protan participant, the sign of $\hat{\beta}_c$ (mechanism class) holds under the PCA-basis loss only. Under the SRM-basis loss the modal argmin is $(32°, 0°)$ in 171 of 300 resamples, leaving $\hat{\beta}_c$ positive in 17% of resamples and negative in 26%, so the two reductions return different mechanism classes for this participant. The deutan sign holds under both bases, at 300 of 300 resamples in each.

S19. Filter-evaluation session: design and comparator

Comparator. The deployed comparator was the macOS accessibility Color Filter (System Settings > Accessibility > Display, build 26.5.1), set at an intensity the participant self-tuned before scanning. It is the unmodified shipping product — an operating-system-level post-display transform. We compared against this deployed filter rather than a re-implemented retinal-simulation transform so that the comparator is the correction an end user actually encounters. The individualized filter, by contrast, was rendered directly in PsychoPy from the frozen per-subject pre-image.

Acquisition. Eight fMRI runs were acquired in ABBA-counterbalanced order, split evenly into four runs per filter. Scanner time set this count: a within-session two-condition design at six runs per condition would run roughly 90 minutes, beyond the 60–75-minute window over which participant attention and head motion stay acceptable. Four runs per condition (∼60 minutes) sits at the recommended boundary.

Run-count adequacy. Four runs retain the Session-1 landmark, the separation between the healthy-control mean and the two CVD cases at hV4. Across all $\binom{6}{n}$ subsets of the six-run Session-1 data, hV4

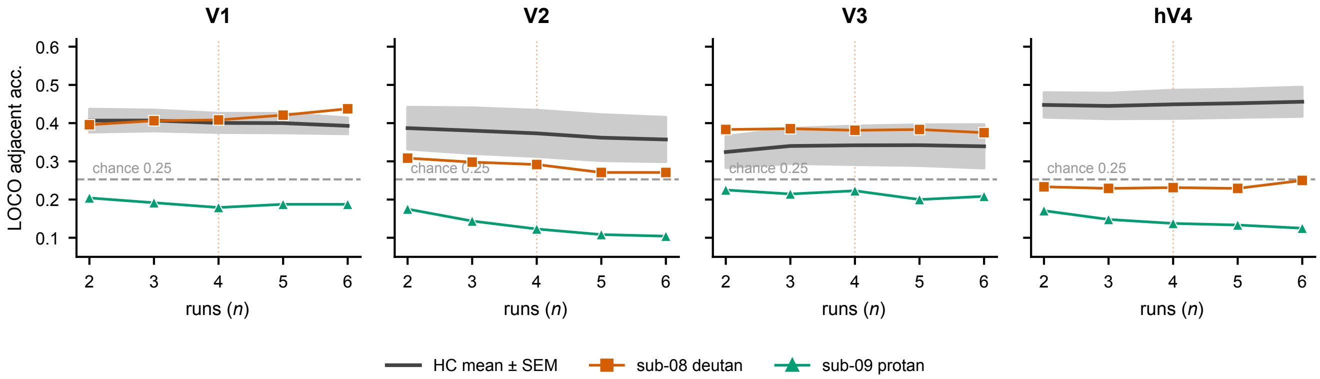

Figure S1: **Hue-interpolation adjacent accuracy as a function of the number of runs.** LOCO adjacent accuracy (canonical FE-6 basis, OLS pseudoinverse decoder) as a function of the number of runs n, computed over all $\binom{6}{n}$ run subsets of the Session-1 data, per ROI. Black line, healthy-control mean $\pm$ SEM (shaded band); dashed line, the 0.25 chance level; markers, the two CVD single cases (sub-08 deutan, sub-09 protan). At hV4 the healthy-control mean stays above chance and the CVD deficit persists at every run count, including the deployed $n = 4$ (orange guide); the metric does not depend on having the full six runs.

LOCO adjacent accuracy held that separation at every run count down to four (Figure S1). At $n = 4$ the healthy-control mean reached 0.45 against 0.46 at six runs and remained above the 0.25 chance level, while both CVD participants stayed below it (deutan 0.23, protan 0.14, single-case $d_{cc} < -2$ in both). A separate subsampling analysis of signal-quality and geometric-stability metrics returned the same ordering, with five runs preferred and four acceptable. The filter conditions each have four runs, whereas the Session-1 HC reference and the unfiltered baseline each have six. Disparity and RDM similarity are inflated by noise in either pattern, so an unmatched comparison penalises the filter conditions twice, once against the baseline and once against the HC distribution that the single-case test uses. Every neural index is therefore reported run-matched. For the interpolation indices the HC reference is subsampled to four runs. For the geometry indices both the HC means and the unfiltered baseline are rebuilt from four runs, exhaustively over all $\binom{6}{4} = 15$ subsets, with the shared response model refitted within each subset and the resulting metrics averaged (`exp2_runmatched_geometry.py`). The filter conditions are never subsampled, since they only ever had four runs. Matching lowers the HC disparity reference from 0.49 to 0.44 at V2 and from 0.45 to 0.43 at V1, and it raises the protan V1 RDM similarity of the unfiltered baseline from 0.25 to 0.33, which moves the individualized filter from above that baseline to below it. The ordering of conditions is otherwise unchanged on all four indices.

Four runs sits below the count recommended for fine-grained RDM-based model comparison. The second session therefore targets movement toward or away from the HC reference rather than adjudication between model classes.

Table S16: **Second-session color classification (LORO eight-way classification accuracy).** Chance $= 0.125$. NF, no-filter (Session-1 baseline); Dep, deployed accessibility filter; Ind, individualized filter. HC, healthy-control mean ($n = 7$).

		Deutan (sub-08)			Protan (sub-09)		
ROI	HC	NF	Dep	Ind	NF	Dep	Ind
V1	0.71	0.79	0.84	0.72	0.79	0.91	0.66
V2	0.71	0.79	0.69	0.62	0.83	0.75	0.72
V3	0.77	0.67	0.78	0.69	0.81	1.00	0.69
hV4	0.75	0.73	0.88	0.50	0.71	0.84	0.69

Single-case inference. Each condition's neural index was compared to the HC distribution as a standardized single-case effect size (Cohen's d against the HC mean and SD), reported in place of an inferential p because the grand-mean permutation null is biased for this design. Robustness was assessed in two ways: by leave-one-run separation (the contrast holding after any single run is dropped) and by re-extracting both conditions on the no-filter baseline voxel set.

Psychophysical battery. Hue-discrimination thresholds were estimated with JND staircases for eight hue pairs, and the eight-alternative forced-choice (8AFC) color-identification task comprised 64 trials. Both were compared to the unfiltered Session-1 baseline and to the HC distribution.

Color classification (LORO). Table S16 reports LORO eight-way classification accuracy for every ROI and condition in both participants. Every cell stayed well above the 0.125 chance level, confirming that the displayed colors remained decodable and that neither filter degraded the second-session color signal.

Forward-tuning encoding index. Figure S2 shows the LOCO forward-tuning correlation ρ to the canonical hue map, a secondary encoding index reported as corroboration only (it lies outside the validated Session-1 decoding pipeline). In the deutan participant the individualized filter reaches the HC level at hV4 ($\rho = +0.18$ vs. HC $+0.21$) while the deployed filter drives it negative ($\rho = -0.39$); in the protan participant the three conditions do not separate (all $\rho \approx -0.02$ at hV4).

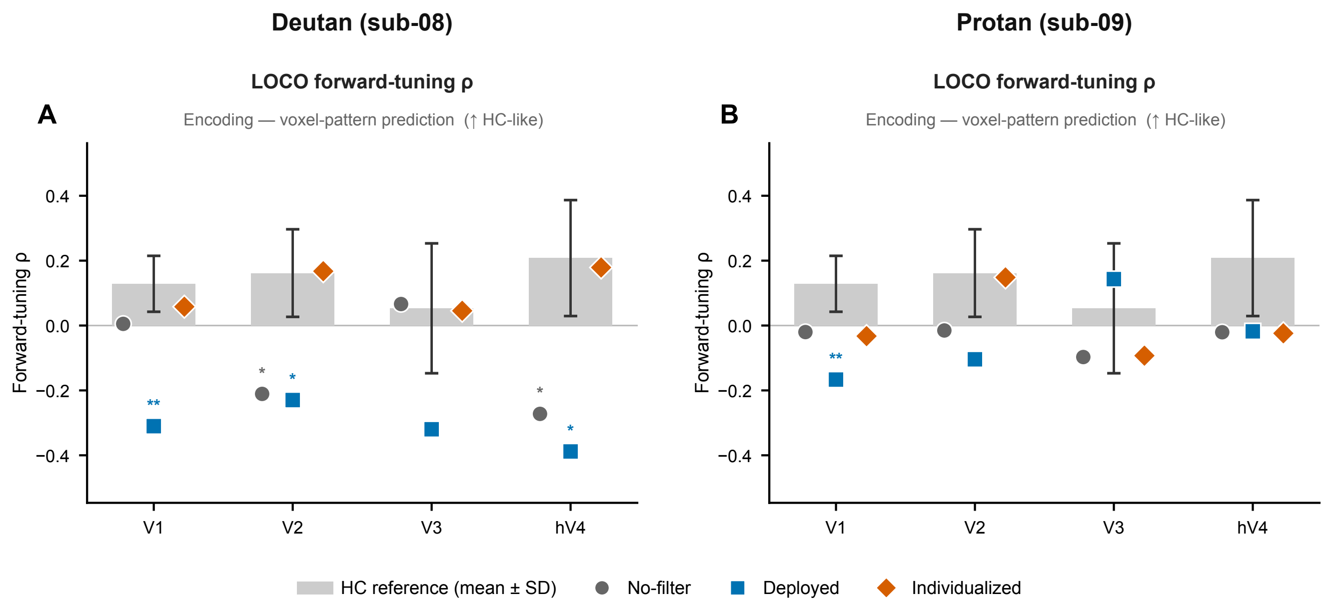

Figure S2: **Forward-tuning encoding index (corroboration only).** LOCO forward-tuning correlation ρ to the canonical hue map across V1–hV4, for the deutan (sub-08, left) and protan (sub-09, right) participants (↑ HC-like). Gray bars, HC reference (mean $\pm$ SD, $n = 7$); markers, no-filter (gray circle), deployed accessibility filter (blue square), individualized filter (orange diamond). Stars, Crawford–Howell single-case deviation from the HC distribution ($^{*}p < .05$, $^{**}p < .01$).

S20. Statistical Analysis

Unless noted otherwise, the significance threshold was $\alpha = 0.05$. False Discovery Rate correction (Benjamini & Hochberg, 1995) was applied within two pre-specified families only: the RDM color-pair comparisons ($q = 0.05$; 28 pairs per participant) and the six parameter-identifiability checks (section 2.13). The exploratory per-hue vulnerability comparisons and the per-pair identification tests (including the blue-hue deficit) are reported one-tailed and uncorrected. Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^{*} = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \tag{8}$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Group-level effect sizes were quantified as Hedges' g (Hedges & Olkin, 1985). Single-case effect sizes are the case-control index d_{cc} defined in §S21. Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S21. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford and Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Effect sizes for the single-case interpolation comparisons are collected in table S17; the geometric-disparity effect sizes are listed alongside their t statistics in table S5 (case-control $d_{cc} = t\sqrt{8/7}$, $n = 7$). Single-case comparisons of within-subject LORO accuracy against the HC distribution reached $|d_{cc}|$ between 0.25 and 1.58 across the eight participant-by-ROI tests, with all $p \geq 0.189$ (two-tailed). The largest deviation was the deutan participant at V3 ($d_{cc} = -1.58$), and at hV4 both participants gave $d_{cc} = -1.08$ ($p = 0.352$).

Cross-subject transfer of the forward encoding model was also compared between destinations. A control-trained model applied to a held-out control reached a mean accuracy of 0.526 over 28 subject-by-ROI cells, and the same model applied to a CVD participant reached 0.432 over 8 cells (Mann–Whitney $U = 163.5$, $p = 0.052$, rank-biserial $r = 0.46$). Both destinations lie far above the 0.125 chance level, and the lowest single cell reached 0.271. This comparison spans participants and is therefore computed in the SRM-aligned space, unlike the within-subject readouts of §S11. sub-10 is excluded here as everywhere else.

Table S17: Effect sizes for the single-case hue-interpolation comparisons at hV4 (main text, §3.2). Crawford–Howell case-control effect size $d_{cc} = t\sqrt{(n+1)/n}$ against the $n = 7$ HC distribution, one-tailed. Per-hue tests are uncorrected and exploratory across eight hues; both CVD participants share identical statistics at the three hues where adjacent accuracy is zero. Blue lies just above the conventional threshold and is the largest per-hue deviation.

Comparison (hV4)	t	p	d_{cc}
LOCO adjacent accuracy, deutan	-1.89	0.054	-2.02
LOCO adjacent accuracy, protan	-3.04	0.012	-3.25
Per-hue adjacent accuracy (one-tailed, exploratory; both participants)			
Blue	-1.93	0.051	-2.06
Purple	-0.79	0.229	-0.85
Magenta	-1.47	0.096	-1.57